BINGHAMTON UNIVERSITY
DEPARTMENT OF MATHEMATICAL SCIENCES

Math 147

Elementary Statistics

Course notes, with every exercise worked out

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About these notes

These are the notes I use in the classroom for Math 147, Elementary Statistics, rewritten so that they can be read away from the blackboard. Nothing has been added for the sake of completeness and nothing has been smoothed over: the order of the topics, the examples and the exercises are the ones we actually do in class.

What is new here is that *every exercise is worked out*. In the original handwritten pages many problems were left for the student, and several solutions were sketched in two lines because I was talking while I wrote them. Here each one is carried out step by step, with the reasoning that justifies the step and not only the arithmetic.

Two conventions are worth stating before we start. A box marked REMEMBER holds something you will use again and again, and is worth memorising. A box marked CAREFUL marks a place where students reliably go wrong; I have put one wherever I have seen the same mistake more than twice.

> **HOW TO USE THIS**
>
> Read the statement, close the book, and try the exercise. Only then read the solution. A solution you have read is not a solution you can produce; statistics is learned with a pencil in your hand.

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CHAPTER 1

Getting Ready for Statistics

Statistics is not an arithmetic course, but it does run on arithmetic. Before we meet a single distribution we need to be fluent with four things: proportions and percentages, rounding, reading a formula, and summation notation. This chapter is that warm-up. Every exercise from the first handout appears here, fully worked.

PROPORTION AND PERCENTAGE

A **proportion** is a fraction $\frac{\#1}{\#2}$, where #2 is the whole (the larger number, the total) and #1 is the part of it you care about. It always lands between 0 and 1. A **percentage** is the same number with the decimal point moved two places to the right:

$$\text{percentage} \;=\; \text{proportion} \times 100\%.$$

The whole difficulty in these problems is deciding what plays the role of #2. Decide that first, and the arithmetic follows.

1.1 Proportions and percentages

Exercise 1

Sage pays \$550 per month for rent and her monthly salary is \$1876.

(a) What proportion of her monthly salary is her rent? (This answer will be between 0.000 and 1.000.)

(b) What percentage of her monthly salary is her rent? (This answer will be between 0.0% and 100.0%.)

Solution. The whole here is the salary, because the rent is a part of it. So #2 $= 1876$ and #1 $= 550$.

(a) The proportion is

$$\frac{550}{1876} = 0.29317\ldots \approx 0.293.$$

(b) Multiplying by 100,

$$0.29317\ldots \times 100 = 29.317\ldots\,\% \approx 29.32\%.$$

Rent eats a little under a third of what she earns.

(a) 0.293 (b) 29.32%

Exercise 2

At Binghamton High School there are 295 seniors who entered Binghamton schools in the fifth grade or lower, and 798 seniors who entered in the sixth grade or higher. What percentage of the seniors entered Binghamton schools in the fifth grade or lower?

Solution. The whole is *all* the seniors, and that number is not given directly: we have to build it.

$$\#2 = 295 + 798 = 1093, \qquad \#1 = 295.$$

Then

$$\frac{295}{1093} = 0.26989\ldots \approx 26.99\%.$$

26.99%

Exercise 3

In a particular arts and humanities college, the following table gives the numbers of men and women in various majors.

	Men	**Women**	**Total**
Anthropology	83	69	152
History	102	110	212
Music	53	37	90
Total	238	216	454

(a) What percentage of the women are anthropology majors?

(b) What percentage of the music majors are men?

Solution. Read each question backwards: the group named *after* the word "of" is the whole.

(a) "of the women" $\Rightarrow$ the whole is all 216 women, and the part is the 69 women in anthropology:

$$\frac{69}{216} = 0.31944\ldots \approx 31.94\%.$$

(b) "of the music majors" $\Rightarrow$ the whole is the 90 music majors, and the part is the 53 men among them:

$$\frac{53}{90} = 0.58888\ldots \approx 58.89\%.$$

(a) 31.94% (b) 58.89%

CAREFUL

Notice how different (a) and (b) are, even though both use numbers from the same table. $\frac{69}{216}$ and $\frac{69}{152}$ answer two different questions: "what percentage of the women study anthropology" and "what percentage of the anthropology students are women".

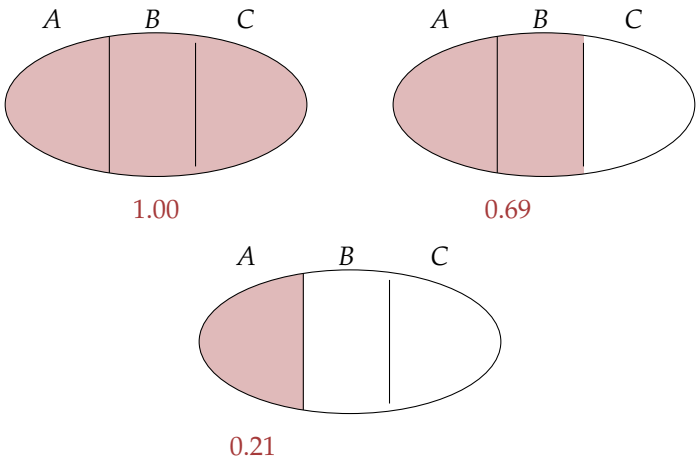
Choosing the wrong denominator is the single most common mistake in this chapter.

1.2 Areas as proportions

Proportions do not have to come from counting people. An area can be a proportion too, and this is exactly how we will read probability later in the course.

Exercise 4

A figure is divided into three non-equal pieces by lines. The three pictures below are all of the same figure with the same division lines. Different areas are shaded and labelled on each. Find the areas of the portions labelled B and C.



Solution. Name the three unknown areas A, B, C. Each picture gives one equation:

$$A + B + C = 1.00, \qquad A + B = 0.69, \qquad A = 0.21.$$

Now subtract, from the smallest piece of information upwards. Taking the first two,

$$C = (A + B + C) - (A + B) = 1.00 - 0.69 = 0.31,$$

and taking the last two,

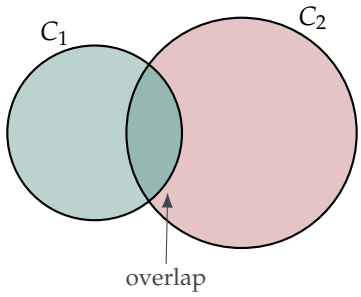
$$B = (A + B) - A = 0.69 - 0.21 = 0.48.$$

As a check, $0.21 + 0.48 + 0.31 = 1.00$, as it must be.

$B = 0.48, \qquad C = 0.31$

Exercise 5

Two circles overlap as sketched below. The area of circle 1 is 0.48 and the area of circle 2 is 0.57.



(a) Can you figure out the shaded area or do you need more information? What would that be?

(b) The overlapping area of the two circles is 0.16. Can you now find the shaded area?

Solution. **(a)** No, not yet. In Exercise 4 the pieces did not overlap, so areas simply added up. Here the two circles share a region, and if we write $0.48 + 0.57$ that shared region has been counted *twice*. What is missing is exactly the area of the overlap.

(b) With the overlap equal to 0.16, count circle 1 without its shared part, and then all of circle 2:

$$\text{shaded} = \underbrace{(0.48 - 0.16)}_{\text{only } C_1} + \underbrace{0.57}_{C_2} = 0.32 + 0.57 = 0.89.$$

Equivalently, add everything and remove the double count: $0.48 + 0.57 - 0.16 = 0.89$.

0.89

OVERLAPPING PIECES

When two regions overlap,

$$\text{area}(C_1 \text{ or } C_2) = \text{area}(C_1) + \text{area}(C_2) - \text{area}(C_1 \text{ and } C_2).$$

This is the addition rule, and we will meet it again word for word when we do probability.

1.3 Rounding and the calculator

HOW TO ROUND

To round to a given decimal place, look at the *next* digit to the right. If it is 5 or more, add one to the digit you are keeping; if it is 4 or less, leave it alone. Then drop everything after it.

Exercise 6

Round 57.39283 to the nearest thousandth.

Solution. "Thousandth" means three decimal places, so we keep 57.392 and look at the fourth decimal:

$$57.392\,\underline{8}3.$$

That digit is $8 \geq 5$, so the third decimal goes up from 2 to 3.

57.393

Exercise 7

Round 0.02397 to the nearest hundredth.

Solution. "Hundredth" means two decimal places, so we keep 0.02 and look at the third decimal:

$$0.02\,\underline{3}97.$$

That digit is $3 \leq 4$, so the second decimal stays as it is. The 97 that follows is irrelevant — only the first discarded digit decides.

0.02

CAREFUL

In Exercise 7 it is tempting to look at the 9 and round up, reasoning that 0.02397 is "almost 0.024". Rounding is decided by one digit only, the first one you drop. Chaining the rounding ($0.02397 \to 0.0240 \to 0.024 \to 0.02$) can give a different answer and is wrong.

Exercise 8

Use your calculator to find $\sqrt{21}$, correct to 3 decimal places.

Solution. The calculator gives $\sqrt{21} = 4.58257569\ldots$. Keeping three decimals we look at the fourth, which is 5, so the third goes up from 2 to 3.

$\sqrt{21} \approx 4.583$

Exercise 9

Use your calculator to find 18^2.

Solution. Squaring means multiplying the number by itself: $18^2 = 18 \times 18 = 324$.

324

Exercise 10

When a scientific calculator shows the answer for a computation as 5.38E − 04, what number does that represent?

Solution. The display 5.38E − 04 is the calculator's way of writing scientific notation:

$$5.38\text{E} - 04 \;=\; 5.38 \times 10^{-4}.$$

A negative exponent moves the decimal point to the *left*, as many places as the exponent says — here four places:

$$5.38 \to 0.538 \to 0.0538 \to 0.00538 \to 0.000538.$$

$5.38\text{E}-04 = 0.000538$

1.4 Inequalities and interval notation

Exercise 11

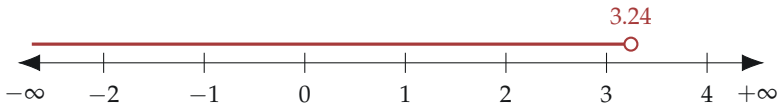
Consider the mathematical statement $x < 3.24$.

(a) How would you read that statement in words?

(b) Give the interval notation for all the numbers that make that statement true. Draw it on a number line.

Solution. **(a)** "x is less than 3.24."

(b) Every number strictly to the left of 3.24 works, and 3.24 itself does not. In interval notation that is $(-\infty, 3.24)$. The round bracket, and the hollow dot in the picture, both say "this endpoint is not included".



$(-\infty, 3.24)$

Exercise 12

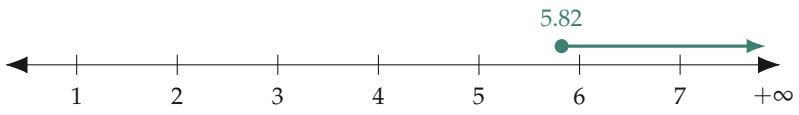
Consider the mathematical statement $y \geq 5.82$.

(a) How would you read that statement in words?

(b) Give the interval notation for all the numbers that make that statement true. Draw it on a number line.

Solution. **(a)** "y is greater than or equal to 5.82."

(b) Now 5.82 *is* allowed, because of the "or equal to". The square bracket, and the solid dot, both say "this endpoint is included".



$[5.82, +\infty)$

BRACKET OR PARENTHESIS

Use a square bracket [ ] when the endpoint belongs to the set ($\leq$ or $\geq$), and a round parenthesis ( ) when it does not ($<$ or $>$). Infinity never belongs to anything, so it always takes a parenthesis.

1.5 Solving for a letter

UNDOING OPERATIONS

To isolate a letter, undo what is being done to it, in reverse order. Each operation is undone by its opposite:

$$+ \longleftrightarrow -, \qquad \times \longleftrightarrow \div, \qquad (\,)^2 \longleftrightarrow \sqrt{\ }.$$

Whatever you do to one side of the equals sign, you do to the other.

Exercise 13

Solve for x: $5 + 3x = 18$.

Solution. The 5 is added, so subtract it from both sides; the 3 multiplies, so divide by it.

$$5 + 3x = 18 \implies 3x = 18 - 5 = 13 \implies x = \frac{13}{3}.$$

As a decimal, $x = 4.333\ldots \approx 4.333$. Check: $5 + 3 \cdot \frac{13}{3} = 5 + 13 = 18$. ✓

$x = \frac{13}{3} \approx 4.333$

Exercise 14

Solve for y: $7y - 8 = 24 + 26y$.

Solution. The letter appears on both sides, so first gather the y's on one side and the numbers on the other.

$$7y - 8 = 24 + 26y \implies 7y - 26y = 24 + 8 \implies -19y = 32 \implies y = -\frac{32}{19}.$$

As a decimal, $y = -1.6842\ldots \approx -1.684$. Check: $7(-1.6842) - 8 = -19.789$ and $24 + 26(-1.6842) = -19.789$. ✓

$y = -\frac{32}{19} \approx -1.684$

Exercise 15

Solve for r: $\dfrac{7}{5} = \dfrac{6}{r}$.

Solution. Two fractions set equal to each other: cross-multiply.

$$\frac{7}{5} = \frac{6}{r} \implies 7r = 5 \cdot 6 = 30 \implies r = \frac{30}{7}.$$

As a decimal, $r = 4.2857\ldots \approx 4.286$.

$r = \frac{30}{7} \approx 4.286$

Exercise 16

Solve for m: $\quad 3.9 = \dfrac{34.2}{m}$.

Solution. Here m sits in a denominator, so first bring it up by multiplying both sides by m:

$$3.9 = \frac{34.2}{m} \implies 3.9\,m = 34.2 \implies m = \frac{34.2}{3.9} = 8.7692\ldots$$

$m = \frac{34.2}{3.9} \approx 8.769$

Exercise 17

Solve for k: $\quad 9.7 = \dfrac{157.4}{\sqrt{k}}$.

Solution. Two steps, in this order: clear the denominator, then undo the square root by squaring.

$$9.7 = \frac{157.4}{\sqrt{k}} \implies 9.7\sqrt{k} = 157.4 \implies \sqrt{k} = \frac{157.4}{9.7} = 16.2268\ldots$$

Now square both sides, using $\left(\sqrt{k}\right)^2 = k$:

$$k = \left(\frac{157.4}{9.7}\right)^2 = (16.2268\ldots)^2 = 263.309\ldots$$

$k = \left(\frac{157.4}{9.7}\right)^2 \approx 263.31$

1.6 Lines

SLOPE-INTERCEPT FORM

$$y = mx + b,$$

where m is the **slope** (the number multiplying x) and b is the **y-intercept** (the number standing alone). This is the simplest model in mathematics, and in this course it is the model we will fit to data.

Exercise 18

For the following equations of lines, find the slope and the y-intercept.

(a) $y = 3x + 8$

(b) $y = 16x - 5.67$

(c) $y = -2 + 5x$

(d) $y = 0.00789 - 1.2345x$

Solution. Match each equation against $y = mx + b$. Watch the *sign* attached to each number, and do not be fooled by the order in which the terms are written.

	equation	slope m	intercept b
(a)	$y = 3x + 8$	3	8
(b)	$y = 16x - 5.67$	16	-5.67
(c)	$y = -2 + 5x$	5	-2
(d)	$y = 0.00789 - 1.2345x$	-1.2345	0.00789

In (c) and (d) the two terms are written in the opposite order. The slope is still whatever multiplies x, and the intercept is still the lone number — position on the page decides nothing.

1.7 Evaluating formulas

In this course the work is more about reasoning than about computation. Even so, you must be able to evaluate a new formula correctly on a basic scientific calculator. A good habit: first run the formula with easy numbers you can check in your head, so that you learn the right order of keystrokes. Only then put in the messy numbers.

Exercise 19

Let $l = x + f \cdot d$.

i. Find l when $f = 3,\ d = 2,\ x = 11$.

ii. Find l when $d = -8,\ f = 3,\ x = 11$.

iii. Find l when $x = 8.79,\ d = 1.7,\ f = 3.23$.

Solution. Multiplication happens before addition, so compute $f \cdot d$ first and only then add x.

i. $l = 11 + 3 \cdot 2 = 11 + 6 = 17.$

ii. $l = 11 + 3 \cdot (-8) = 11 - 24 = -13.$

iii. $l = 8.79 + 3.23 \cdot 1.7 = 8.79 + 5.491 = 14.281.$

In (ii) the product is negative, so adding it moves the answer down. If your calculator returned -104 you typed $(11 + 3) \cdot (-8)$: the parentheses were not there.

i. 17 ii. -13 iii. 14.281

Exercise 20

Let $w = \dfrac{x - m}{s/\sqrt{n}}$.

i. Find w when $x = 9,\ m = 6,\ s = 8,\ n = 16$.

ii. Find w when $x = 0.49,\ m = 0.5,\ s = 0.063,\ n = 17$.

Solution. Build the numerator and the denominator separately, then divide once at the end.

i. Numerator $9 - 6 = 3$. For the denominator, $\sqrt{16} = 4$, so $s/\sqrt{n} = 8/4 = 2$. Hence

$$w = \frac{3}{2} = 1.5.$$

ii. Numerator $0.49 - 0.5 = -0.01$, which is negative — that is fine and it means x sits below m. For the denominator, $\sqrt{17} = 4.12310\ldots$, so

$$\frac{s}{\sqrt{n}} = \frac{0.063}{4.12310\ldots} = 0.0152798\ldots$$

and therefore

$$w = \frac{-0.01}{0.0152798\ldots} = -0.65446\ldots \approx -0.654.$$

i. $w = 1.5$ ii. $w \approx -0.654$

CAREFUL

This w is not a random formula: it is the z-statistic we will use all term. Get the keystrokes right now. On most calculators you must write the denominator inside parentheses, `(0.063/\17)`, or the machine will divide only by 0.063 and then multiply by $\sqrt{17}$.

Exercise 21

Let $R = \sqrt{\dfrac{m_1^2}{n_1} + \dfrac{m_2^2}{n_2}}$.

i. Find R when $m_1 = 6,\ n_1 = 4,\ m_2 = 8,\ n_2 = 4$.

ii. Find R when $m_1 = 6.21,\ n_1 = 32,\ m_2 = 8.73,\ n_2 = 49$.

Solution. Everything under the root must be finished before the root is taken.

i.

$$\frac{6^2}{4} + \frac{8^2}{4} = \frac{36}{4} + \frac{64}{4} = 9 + 16 = 25, \qquad R = \sqrt{25} = 5.$$

ii.

$$\frac{6.21^2}{32} = \frac{38.5641}{32} = 1.205128\ldots, \qquad \frac{8.73^2}{49} = \frac{76.2129}{49} = 1.555365\ldots$$

Adding, $1.205128 + 1.555365 = 2.760493\ldots$, and then

$$R = \sqrt{2.760493\ldots} = 1.66147\ldots \approx 1.661.$$

i. $R = 5$ ii. $R \approx 1.661$

CAREFUL

A square root does *not* distribute over a sum:

$$\sqrt{x+y} \neq \sqrt{x} + \sqrt{y}.$$

In part (i), $\sqrt{9+16} = \sqrt{25} = 5$, whereas $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$. Different answers. Add first, then take the root.

Exercise 22

Let $r = \dfrac{m-t}{s}$.

i. Find m when $r = 2,\ s = 3,\ t = 4$.

ii. Find m when $r = 1.84,\ s = 3.12,\ t = 3.1$.

Solution. This time the unknown is not the subject of the formula, so rearrange first. Multiply both sides by s and then move t across:

$$r = \frac{m-t}{s} \implies rs = m - t \implies m = rs + t.$$

Only now substitute.

i. $m = 2 \cdot 3 + 4 = 6 + 4 = 10.$

ii. $m = 1.84 \cdot 3.12 + 3.1 = 5.7408 + 3.1 = 8.8408 \approx 8.841.$

i. $m = 10$ ii. $m \approx 8.841$

1.8 Summation notation

The notation $\sum_{i=1}^{n} x_i$ stands for the sum of n numbers $x_1, x_2, \ldots, x_n$, and is read *the sum of all* x_i *with* i *running from* 1 *to* n:

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + \cdots + x_n.$$

The term following the sign $\sum$ (the Greek letter sigma) says *what* is being added; the labels below and above say *how many* terms there are.

Exercise 23

Let x represent the number of M&M's contained in individual trail mix snacks. The data for five trail mix snacks is recorded below.

Trail mix (i)	1	2	3	4	5
M&M's count (x_i)	6	5	6	7	8

(a) The formula that means to sum up the M&M's counts of all five trail mixes is $\sum_{i=1}^{5} x_i$. Evaluate it.

(b) Show the formula that means to sum up the M&M's counts of the first three trail mixes, and evaluate it.

(c) The formula that means to sum up the squares of all M&M's counts is $\sum_{i=1}^{5} x_i^2$. Evaluate it.

(d) The formula for the square of the sum of all M&M's counts is $\left(\sum_{i=1}^{5} x_i\right)^2$. Evaluate it.

(e) Fill in the blank with the proper symbol, $=$ or $\neq$: $\displaystyle\sum_{i=1}^{5} x_i^2 __ \left(\sum_{i=1}^{5} x_i\right)^2$.

Solution. **(a)** Add the five counts:

$$\sum_{i=1}^{5} x_i = x_1 + x_2 + x_3 + x_4 + x_5 = 6 + 5 + 6 + 7 + 8 = 32.$$

(b) "The first three trail mixes" means stopping at $i = 3$. What is being added is still x_i, the counts — not the labels i:

$$\sum_{i=1}^{3} x_i = x_1 + x_2 + x_3 = 6 + 5 + 6 = 17.$$

(c) Now each count is squared *before* adding:

$$\sum_{i=1}^{5} x_i^2 = 6^2 + 5^2 + 6^2 + 7^2 + 8^2 = 36 + 25 + 36 + 49 + 64 = 210.$$

(d) Here the adding happens first and the squaring last. Using the total from (a),

$$\left(\sum_{i=1}^{5} x_i\right)^2 = 32^2 = 1024.$$

(e) Comparing (c) and (d), 210 against 1024:

$$\sum_{i=1}^{5} x_i^2 \neq \left(\sum_{i=1}^{5} x_i\right)^2.$$

(a) 32 (b) 17 (c) 210 (d) 1024 (e) $\neq$

CAREFUL

Parts (c) and (d) look almost the same written down, and they are not the same number at all: the sum of the squares is 210, the square of the sum is 1024. Square-then-add and add-then-square are different instructions. Both appear in the formula for the standard deviation, so the distinction will cost you marks if you let it slide now.

CHAPTER 2

Displaying and Describing Data

2.1 What statistics is about

Three ideas run underneath everything we will do this term.

> **THE THREE IDEAS**
>
> **Variation.** Data vary, because we do not see everything, and even what we do see we measure imperfectly.
> **Learning from the data.** We hope to learn about the world as well as we can from the limited, imperfect data we have.
> **Making intelligent decisions.** The better we understand the world, the wiser our decisions will be.

Notice that the first idea is not an apology. Variation is not a defect in the data to be removed; it is the subject matter. Statistics is the discipline of saying something reliable in spite of it, and of being honest about how reliable.

2.2 The *Who* and the *What*

In general the rows of a data table correspond to individual **cases**, about whom — or about which, if they are not people — we record some characteristics. Cases go by different names depending on the situation:

- individuals who answer a survey are **respondents**;
- people on whom we experiment are **subjects**, or **participants**;
- animals, plants, websites and other inanimate subjects are often called **experimental units**;
- in a database the rows are **records**;
- often we simply call cases what they are: customers, economic quarters, companies.

Whatever we call them, the rows are the *Who* of the data.

The characteristics recorded about each individual are the **variables**. They are usually the columns of the data table, with a name in the header saying what has been recorded. In a table of music orders, the variables are the order number, the name, the state or country, the price, the area code, and so on.

Categorical and quantitative variables

Some variables just tell us what group or category each individual belongs to. Are you male or female? Pierced or not? We call these **categorical** (or qualitative) variables. Their values

are often text — the variable *state/country* is clearly categorical — but *numerals are often used as labels*, so a categorical variable can perfectly well look numeric. An area code such as 13905 in Binghamton is a categorical variable: no one adds or averages area codes.

When a variable contains measured numerical values with units, we call it a **quantitative** variable.

Exercise 1 (Consumer Reports tablets)

A *Consumer Reports* article about 126 tablets lists each tablet's manufacturer, price, battery life (hrs.), the operating system (Android, iOS or Windows), an overall quality score (0–100), and whether or not it has a memory card reader. Are these variables categorical or quantitative? Include units where appropriate, and describe the *Why* of this investigation.

Solution. Ask of each one: does it name a group, or does it measure an amount with units?

variable	type	units
manufacturer	categorical	—
price	quantitative	dollars
battery life	quantitative	hours
operating system	categorical	—
quality score	quantitative	none (a 0–100 scale)
memory card reader	categorical	—

The *Who* is the sample of 126 tablets; each one selected represents all similar tablets offered by that manufacturer. The *Why* is that the magazine hopes to give consumers the information they need to choose a good tablet.

CAREFUL

The quality score is quantitative even though it has no units, because it measures an amount on a scale where larger means more. The operating system is categorical even if someone codes Android as 1, iOS as 2 and Windows as 3. Type is decided by what the variable *means*, never by whether it looks like a number.

2.3 How the data are collected

How the data are collected can make the difference between insight and nonsense. We will require, from here on, that the data be collected **randomly** — or at least that we are willing to assume so. Later we will study that random condition properly, with probability and with models for distributions.

CAREFUL

The random condition is not a formality. Without it we can reach confident, well-computed, completely wrong conclusions. A survey of who likes a colour of shirt, taken among people already wearing it, produces excellent arithmetic about nothing.

2.4 Models

A **model** is a summary and a simplification of the data, which helps our understanding in many ways. It is a simplification of reality that gives us information we can learn from and use, even though it does not represent reality exactly.

Kepler's laws are the great example. From Tycho Brahe's observations of the declination of Mars over the twenty years before 1600 — a cloud of points with gaps in it — Kepler saw through the small anomalies of measurement and produced three simple laws. The value of the model is visible exactly where the data are not: for a year such as 1598 there are no observations, and yet the curve gives us a declination of about 25 degrees for it. With a model we obtain a great deal of information about our data, and we can also ask about the future.

2.5 Frequency tables and bar charts

Categorical variables are easy to summarize in a **frequency table**, which lists the number of cases in each category along with its name. A **relative frequency table** displays percentages, or proportions, instead of counts. Both show the **distribution** of a categorical variable, because both name the possible categories and say how often each one occurs.

> **Exercise 2** (Titanic ticket class)
>
> Among the people aboard the *Titanic*, the categorical variable *ticket class* had these counts: first class 324, second class 285, third class 710, crew 889. Build the frequency table and the relative frequency table.

Solution. The counts are given, so the frequency table is immediate. For the relative frequency table, every percentage is $\frac{\#1}{\text{total}}$, and the total must be built first:

$$\text{total} = 324 + 285 + 710 + 889 = 2208.$$

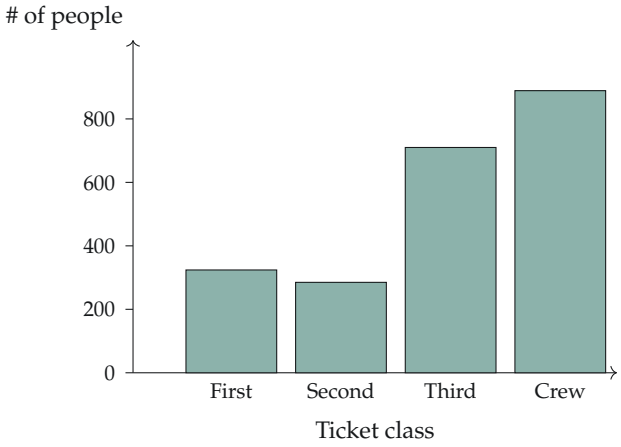
Then, one category at a time,

$$\frac{324}{2208} = 0.1467\ldots = 14.67\%, \qquad \frac{285}{2208} = 12.91\%,$$
$$\frac{710}{2208} = 32.16\%, \qquad \frac{889}{2208} = 40.26\%.$$

class	count	percentage
First	324	14.67%
Second	285	12.91%
Third	710	32.16%
Crew	889	40.26%
Total	2208	100.00%

Here the percentages add to exactly 100%, but they need not: rounding each one separately can leave the sum a little above or below 100.

A **bar chart** displays the same distribution as bars, one per category, with gaps between them, because the categories have no order of their own.



Now it is easy to see that the majority of the people on board were not crew.

2.6 Histograms

We cannot make a bar chart for a quantitative variable, because quantitative variables have no categories. Instead we make a **histogram**.

> **BAR CHART OR HISTOGRAM**
>
> Both use bars, and they are fundamentally different.
> In a **bar chart** the horizontal axis just names categories. The bars could be put in any order, and they are drawn with a space between them.
> In a **histogram** the horizontal axis shows the values of the variable *in order*. The axis is sliced into equal-width **bins**, and the bars show the count in each bin. Here the bars touch, and a **gap** is meaningful: it marks a region where there are no observations at all.

In the histogram of the ages of those aboard the *Titanic*, each bin is 5 years wide, so the tallest bar says that the most populous age group was the 20 to 24 year olds, with over 400 people. The youngest passengers were infants and the oldest was more than 70 years old.

2.7 Shape

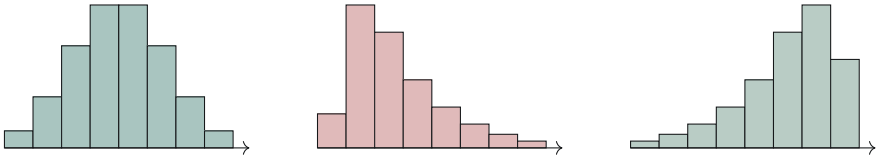
We summarize the shape of a distribution in terms of three properties: how many modes it has, whether it is symmetric or skewed, and whether it has any extraordinary cases, or outliers.

Modes

A hump in the histogram is called a **mode**. For a categorical variable the mode is simply the most frequent category; for a quantitative variable it makes more sense to use the word for the *peak* of the histogram than for a single value. One peak is **unimodal**, two peaks **bimodal**, three or more **multimodal**. A histogram with no mode at all, in which the bars are all about the same height, is called **uniform**.

Symmetry and skewness

A distribution is **symmetric** if you could fold the histogram along a vertical line through the middle and have the edges match closely. The thinner ends of a distribution are its **tails**, and if one tail stretches out farther than the other the histogram is said to be **skewed** to the side of the longer tail.



symmetric skewed to the right skewed to the left

We generally prefer symmetric distributions, because they are easier to summarize, to model and to discuss: it is pretty clear where the centre is.

Outliers and gaps

Always mention any stragglers, or **outliers**, that stand away from the body of the distribution. If you were collecting data on nose lengths and Pinocchio were in the group, he would be an outlier, and you would certainly want to mention him.

> **WHAT TO DO WITH AN OUTLIER**
>
> An outlier can be the most informative part of your data. Or it might just be an error. *Never throw data away without comment.* Treat outliers specially and discuss them when you tell about your data; or find the error and fix it if you can. Be sure to look for outliers — always.

You should also notice any **gaps**, where no data appear at all. Gaps may indicate separate modes, or even separate subgroups worth considering individually. But do not be overzealous: in a small data set a gap may be pure happenstance.

> **Exercise 3** (Credit card expenditures)
>
> A credit card company provides data on the amount spent by 500 selected customers during a three-month period. The histogram shows a very tall bar near zero, a long thin tail to the right reaching an extraordinarily large value at about $7000, and a few values below zero. Describe the shape of this distribution.

Solution. Answer the three questions of shape, in order.

Modes: one tall peak at the low end, so the distribution is unimodal.

Symmetry: the right tail is very much longer than the left, so it is skewed to the high end — skewed to the right.

Unusual features: there is an extraordinarily large value at about $7000, which should be mentioned and investigated rather than deleted. Some of the expenditures are negative, which for a credit card is not an error at all: a refund or a credit produces a negative charge.

2.8 The centre: median and mean

The median

One natural choice of typical value is the **median**: the value literally in the middle, with half the values below it and half above. The median has the same units as the data — say so whenever you report one. The median age of crew members on the *Titanic* was 30 years.

FINDING THE MEDIAN BY HAND

Order the values first; everything else depends on it.

If n is **odd**, the median is the single middle value, in position $\dfrac{n+1}{2}$.

If n is **even**, there are two middle values, in positions $\dfrac{n}{2}$ and $\dfrac{n}{2}+1$, and the median is their average.

Exercise 4 (Median by hand)

Find the median of

$$14.1, \quad 3.2, \quad 25.3, \quad 2.8, \quad -17.5, \quad 13.9, \quad 45.8,$$

and then of the same batch with one more value, 35.7, added.

Solution. **First batch.** Order the values:

$$-17.5,\ 2.8,\ 3.2,\ \mathbf{13.9},\ 14.1,\ 25.3,\ 45.8.$$

There are $n = 7$ values, an odd number, so the median sits in position $\frac{7+1}{2} = 4$. Counting in from either end, that is 13.9. Three values are lower and three are higher, as it should be.

Second batch. With 35.7 added the ordered values are

$$-17.5,\ 2.8,\ 3.2,\ \mathbf{13.9},\ \mathbf{14.1},\ 25.3,\ 35.7,\ 45.8.$$

Now $n = 8$ is even, so we average the values in positions $\frac{8}{2} = 4$ and $\frac{8}{2}+1 = 5$:

$$\text{median} = \frac{13.9 + 14.1}{2} = 14.0.$$

Four data values are lower and four are higher.

13.9 and 14.0

The mean

The most common way to summarize the centre of a distribution is with the **mean**:

$$\bar{y} = \frac{\text{total}}{n} = \frac{\sum y_i}{n},$$

where n is the sample size, that is, the number of data values.

The mean is located at the *balancing point* of the histogram — of the histogram, not of a bar chart. Picture the histogram cut out of stiff card: the mean is where it would balance on a knife edge.

Which one, the mean or the median?

Both summarize the centre, and they do it in different ways. The median is *less influenced by skewness or by outliers* than the mean, which makes it a good choice for skewed distributions such as income, or company size, or the price of a ticket.

The *Titanic* fares show this clearly. The distribution of ticket prices is skewed to the right; the mean is £33.0 while the median is £14.4. The handful of very expensive tickets at the right have pulled the mean towards them and away from the median. The mean is not the middle point.

> **CHOOSING A CENTRE**
>
> For a **skewed** distribution, report the **median**.
> For a **symmetric** distribution, the **mean** is the better summary.
> When the two differ a lot, that difference is itself information: it is telling you the distribution is skewed.

2.9 The spread

We also want to measure the **spread**: how much the data values vary around the centre. When you describe a distribution numerically, always report a measure of its spread along with its centre. Spread, like shape and centre, is a general description of the distribution, not a precise quantity.

The range

$$\text{Range} = \max - \min\,.$$

The range is a single number, not an interval, despite the way the word is used in ordinary speech. The maximum *Titanic* crew age was 62 and the minimum was 14 — the bellboy — so the range is $62 - 14 = 48$ years.

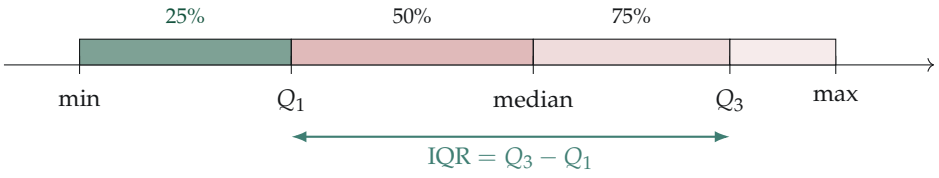
Its weakness is that one extreme value can make it enormous and unrepresentative, so the range should never be your only measure of spread.

Quartiles and the interquartile range

The value just above the lower 25% of the data is the **lower quartile**, or first quartile, or 25th percentile, written Q_1. The **upper quartile** Q_3 is the 75th percentile. The median is the 50th percentile. The distance between the quartiles is the **interquartile range**:

$$\text{IQR} = Q_3 - Q_1,$$

the range of just the middle half of the data.



For the *Titanic* crew ages the median is 30 years: ordering the 889 ages, the 445th is 30. Taking the lower 445 ages, their median is 24, and that is Q_1; in the upper half the median is 37, and that is Q_3. So

$$\text{IQR} = 37 - 24 = 13 \text{ years}.$$

The range is 48 years and the middle half covers only 13: that contrast is the useful part.

> **Exercise 5** (Spread of the credit card expenditures)
>
> The distribution of credit card expenditures is skewed, and the quartiles are \$73.84 and \$624.80. What is the IQR, and why is it a suitable measure of spread here?

Solution.

$$\text{IQR} = 624.80 - 73.84 = 550.96 \text{ dollars}.$$

It is suitable for the same reason the median was: like the median, the IQR is built from positions in the ordered data rather than from the values themselves, so it is not affected by the outlying \$7000 charge or by the skewness. The range, by contrast, would be dominated by that single customer.

IQR = \$550.96

2.10 Variance and standard deviation

The IQR is almost always a reasonable summary, but because it uses only the two quartiles it ignores much of the information about how the individual values vary. The **standard deviation** takes into account how far *each* value is from the mean.

We first define the **variance**:

$$s^2 = \frac{\sum (y_i - \bar{y})^2}{n - 1},$$

where $\bar{y}$ is the mean, n the sample size and y_i each data value.

> **Exercise 6** (Variance and standard deviation by hand)
>
> For the data $1,\ 3,\ 7,\ 22,\ 55$, compute the mean, the variance and the standard deviation.

Solution. Here $n = 5$ and $y_1 = 1, y_2 = 3, y_3 = 7, y_4 = 22, y_5 = 55$.

The mean.

$$\bar{y} = \frac{\sum y_i}{n} = \frac{1 + 3 + 7 + 22 + 55}{5} = \frac{88}{5} = 17.6.$$

The deviations and their squares. Subtract the mean from each value and square; it is worth laying this out in a table, because that is where the arithmetic goes wrong.

y_i	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
1	-16.6	275.56
3	-14.6	213.16
7	-10.6	112.36
22	4.4	19.36
55	37.4	1398.76
		2019.20

The variance. Divide by $n - 1 = 4$, not by n:

$$s^2 = \frac{2019.20}{4} = 504.80.$$

The standard deviation. The variance is in squared units, which is why we take the square root:

$$s = \sqrt{s^2} = \sqrt{504.80} = 22.467\ldots \approx 22.47.$$

$\bar{y} = 17.6, \quad s^2 = 504.80, \quad s \approx 22.47$

CAREFUL

Two places to be careful. First, the denominator is $n - 1$, not n: dividing by 5 here would give 403.84, which is not the sample variance. Second, the deviations must be squared *before* they are added — they sum to zero otherwise, every time, which is exactly why we square them.

UNITS

The variance is in squared units: squared dollars, squared years, mpg^2. Nobody wants to discuss squared ages, and that is the whole reason the standard deviation exists:

$$s = \sqrt{\frac{\sum(y - \bar{y})^2}{n - 1}}$$

is back in the units of the data. In practice we almost always let technology compute s, but you should be able to do it by hand once, as above, so that you know what the machine is doing.

2.11 Putting it together

Exercise 7 (Question for you: the five-number summary)

How do we compute Q_1, the median and Q_3 from a set of data? Do it for 1, 3, 7, 22, 55, and give the range and the IQR as well.

Solution. **Step 1: order the data.** They already are: 1, 3, 7, 22, 55.

Step 2: the median. Here $n = 5$ is odd, so the median is the value in position $\frac{5+1}{2} = 3$, that is

$$\text{median} = 7.$$

Step 3: the quartiles. Q_1 is the median of the lower half and Q_3 the median of the upper half. With an odd n the median itself is *not* included in either half, so

$$\text{lower half} = \{1,\ 3\} \implies Q_1 = \frac{1+3}{2} = 2,$$
$$\text{upper half} = \{22,\ 55\} \implies Q_3 = \frac{22+55}{2} = 38.5.$$

Step 4: range and IQR.

$$\text{Range} = 55 - 1 = 54, \qquad \text{IQR} = 38.5 - 2 = 36.5.$$

The five-number summary is therefore

$$\min = 1, \quad Q_1 = 2, \quad \text{median} = 7, \quad Q_3 = 38.5, \quad \max = 55.$$

Compare this with the mean, 17.6, computed in the previous exercise. The mean is more than twice the median, which tells us at once — without drawing anything — that this little batch is strongly skewed to the right, pulled by the 55.

$Q_1 = 2, \quad \text{median} = 7, \quad Q_3 = 38.5, \quad \text{IQR} = 36.5$

THE REPORTING HABIT

Whenever you describe a distribution, say three things and in this order: **shape**, then **centre**, then **spread** — and let the shape decide the other two. Skewed: median and IQR. Symmetric: mean and standard deviation. Then mention any outliers separately, by name.

CHAPTER 3

The Standard Deviation as a Ruler and the Normal Model

3.1 Choosing the right summaries

We now have two pairs of summaries, and the choice between them is not a matter of taste. It is decided by the shape of the distribution.

Exercise 1 (Choosing summary statistics)

You have given a credit card company's board of directors a histogram of customer expenditures and summarized the centre and spread with the median and the IQR. Knowing a little statistics, the directors now insist on the mean and the standard deviation instead. You calculate that the mean is \$478.19 and the standard deviation is \$741.87. But you know from the histogram that there is an outlier at \$7000. Explain to the directors why the mean and standard deviation should be interpreted differently from the median and IQR.

Solution. Two things go wrong at once, and it is worth separating them.

The mean is dragged. The high outlier at \$7000 pulls the mean upward substantially. If you locate \$478.19 on the histogram you will see it sitting out in the thin part of the distribution, where almost nobody is: it is not a typical value at all.

The standard deviation is inflated. The standard deviation is built from the squared deviations $(y_i - \bar{y})^2$. For the outlier that deviation is enormous, and squaring makes it enormous again, so one single customer contributes more to s^2 than hundreds of ordinary customers together. The reported \$741.87 therefore suggests that expenditures vary far more than they really do — note that it is larger than the mean itself.

The median and the IQR are built from *positions* in the ordered data, not from the values, so a single extreme value moves them hardly at all. For skewed data with outliers they give a more realistic description of centre and spread.

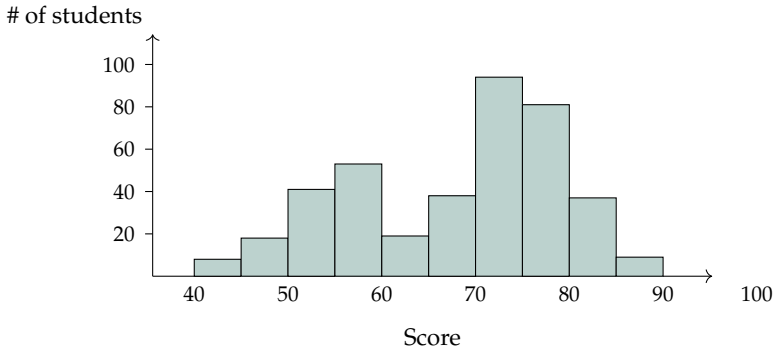
WHICH PAIR OF SUMMARIES

Symmetric distribution $\longrightarrow$ **mean** (centre) and **standard deviation** (spread).
Skewed distribution $\longrightarrow$ **median** (centre) and **IQR** (spread).
So the first question is never "what is the average?". It is "what shape is this?". Identify symmetric or skewed first; the right summaries follow.

3.2 A complete example: a class with two groups

Exercise 2 (Test scores in a large class)

Test scores from a large calculus class of 400 students are shown in the histogram below.



(a) Describe the distribution of scores. What might account for this shape?

(b) Why might both the mean and the median score be misleading as a summary of the centre?

Solution. **(a)** The distribution is *not* symmetric. It appears to be **bimodal**: there are two distinct peaks, one around the scores 60–65 and a second, higher one around 75–80. There is also a slight skew to the left, since there are more students with high scores than with low ones.

A bimodal shape is a signal, not a nuisance. It suggests the class is really made up of *two different groups* of students — one that scored high and one that scored low. Study habits, attendance, or preparation coming into the course are all plausible reasons for the split.

(b) Both the mean and the median will land somewhere between the two peaks, probably near a score of 70 — which is precisely where comparatively few students actually are. That midpoint does not describe either group.

Remember what a centre is *for*: it should let someone who has not seen the histogram picture the data. Here a single number hides the two facts that matter, namely that a large group scored significantly higher and another group significantly lower. Summarizing these data with one central value is misleading. The right move is to study the two groups separately.

WHEN ONE NUMBER IS THE WRONG ANSWER

A centre and a spread describe a distribution well only when it has *one* centre. If the histogram is bimodal, say so, and split the data before summarizing.

3.3 Standardizing: z-scores

Often we need to compare two values that are measured on different scales: seconds against seconds in two different races, or a test score against a height. The standard deviation gives us the ruler to do it with, because it takes into account how far each value is from the mean.

Expressing a distance from the mean *in standard deviations* **standardizes** the value. To standardize, subtract the mean and divide the difference by the standard deviation:

$$z = \frac{y - \bar{y}}{s}.$$

We call these **z-scores**.

READING A z-SCORE

A z-score measures the distance of a value from the mean, counted in standard deviations. $z = 2$ means the value is 2 standard deviations *above* the mean.
Values below the mean have *negative* z-scores, so $z = -1.6$ means the value was 1.6 standard deviations below the mean.
z-scores have no units. That is exactly what makes them comparable across different variables.

Exercise 3 (Standardizing skiing times)

The men's super combined skiing event consists of two races, a downhill and a slalom; the times are added and the lowest total wins. At the 2010 Vancouver Olympics the mean slalom time was 52.67 seconds with a standard deviation of 1.614 seconds, and the mean downhill time was 116.26 seconds with a standard deviation of 1.914 seconds. Bode Miller won the gold with a combined time of 164.92 seconds, skiing the slalom in 51.01 and the downhill in 113.91. On which race did he do better compared with the competition?

Solution. The two races cannot be compared directly: the downhill is a much longer race, so its times are bigger and vary differently. Standardize each one.

$$z_{\text{slalom}} = \frac{y - \bar{y}}{s} = \frac{51.01 - 52.67}{1.614} = \frac{-1.66}{1.614} = -1.03,$$

$$z_{\text{downhill}} = \frac{113.91 - 116.26}{1.914} = \frac{-2.35}{1.914} = -1.23.$$

Now the interpretation, and here is the trap: in a race, *faster times are smaller*, so a good performance gives a *negative* z-score. The more negative, the better.

His downhill time was 1.23 standard deviations below the mean, against 1.03 for the slalom. So his downhill was the more remarkable of the two.

the downhill: $z = -1.23$ against $z = -1.03$

CAREFUL

Do not read "higher z means better". A z-score says how far from the average a value is, and whether far-above or far-below is good depends entirely on the variable. Low is good for race times, high is good for exam scores.

Exercise 4 (Combining z-scores)

Ted Ligety had times of 50.76 seconds in the slalom — the fastest of the 35 finalists — and 115.06 seconds in the downhill, a total of 165.82 seconds, almost a second behind Miller, leaving him fifth. The downhill is longer than the slalom and so counts for more in the total. Would the placement have changed if each event had been treated equally, by standardizing each and adding the standardized scores?

Solution. Miller's z-scores were -1.03 and -1.23, so his total is

$$-1.03 + (-1.23) = -2.26.$$

For Ligety,

$$z_{\text{slalom}} = \frac{50.76 - 52.67}{1.614} = \frac{-1.91}{1.614} = -1.18, \qquad z_{\text{downhill}} = \frac{115.06 - 116.26}{1.914} = \frac{-1.20}{1.914} = -0.63,$$

and his total is

$$-1.18 + (-0.63) = -1.81.$$

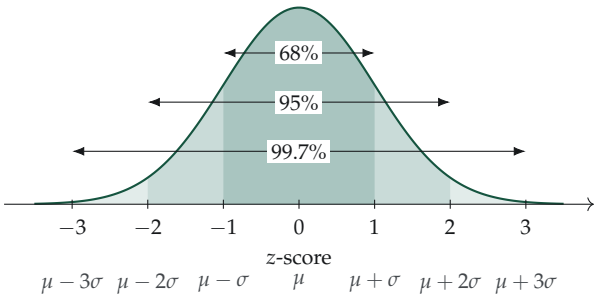
Since -2.26 is further below zero than -1.81, Miller still comes out ahead. Ligety beat Miller in the slalom, but Miller beat him by *more* in the downhill. Using standardized scores would not have put Ligety ahead.

No: -2.26 (Miller) beats -1.81 (Ligety)

3.4 Normal models and the 68–95–99.7 Rule

In 1733 the French mathematician Abraham de Moivre showed that for many unimodal and symmetric distributions, about 68% of the values fall within one standard deviation of the mean, about 95% within two standard deviations, and about 99.7% — virtually all of them — within three. We call this the **68–95–99.7 Rule**.

In 1809 Gauss worked out the formula for the model that accounts for de Moivre's observation. That model is called the **Normal**, or Gaussian, model.



Parameters and statistics

The picture above is drawn in z-scores, but we could just as easily put the mean and the standard deviation back in. When we do, we write $N(\mu, \sigma)$ for a Normal model with mean μ and standard deviation σ.

Why the Greek letters? Because μ and σ are *not* numerical summaries of data. They are characteristics of the *model*, called **parameters**: the values we choose in order to specify the model completely. Summaries computed from data — $\bar{y}$, the median, s — are called **statistics**, and are written with Latin letters.

GREEK OR LATIN

$$\underbrace{\mu,\ \sigma}_{\text{parameters: the model}} \qquad \text{versus} \qquad \underbrace{\bar{y},\ s}_{\text{statistics: the data}}$$

The whole of inference, later in the course, is about using the Latin letters to say something about the Greek ones.

If we model data with a Normal model and standardize using the corresponding μ and σ, the standardized value is still called a z-score:

$$z = \frac{y - \mu}{\sigma}.$$

Once the data are standardized we need only the model $N(0, 1)$, with mean 0 and standard deviation 1. That one is called the **standard Normal model**.

3.5 Using the rule

Exercise 5 (Wrist circumference)

A study of men's health measured 14 body characteristics of 250 men, one of which was wrist circumference, in cm. The mean wrist circumference was 18.22 cm and the standard deviation was 0.91 cm; the histogram of the 250 values is unimodal and symmetric. Will the 68–95–99.7 Rule give useful percentages here? About how many of these men does the rule predict will have a wrist circumference larger than 20.04 cm?

Solution. **Is the rule applicable?** Yes. The distribution is unimodal and symmetric, which is the condition the rule needs.

Locate 20.04 **on the ruler.** Standardize:

$$z = \frac{20.04 - 18.22}{0.91} = \frac{1.82}{0.91} = 2,$$

so 20.04 cm is exactly two standard deviations above the mean.

Apply the rule. About 95% of the values lie within two standard deviations, which leaves 5% outside. By symmetry that 5% splits evenly between the two tails, so about 2.5% lies above $z = 2$.

Turn the percentage into a count.

$$2.5\% \text{ of } 250 = 0.025 \times 250 = 6.25 \approx 6 \text{ men.}$$

Which is, as it happens, exactly the number of men in the data set with wrists larger than 20.04 cm.

about 6 men

CAREFUL

The step that students skip is the halving. "95% within two SDs" leaves 5% outside, but the question asked only about the *upper* tail, so the answer is 2.5%, not 5%. Draw the curve and shade the piece you were asked about before you compute anything.

3.6 When the rule does not apply

Exercise 6 (CEO compensation)

The total compensation of the CEOs of 434 top public companies in 2014 has a histogram that is strongly skewed to the right, with a long thin tail reaching past \$150M. Will the 68–95–99.7 Rule give useful percentages here? Using the rule, about how many of these CEOs should have compensations within one standard deviation of the mean, that is, between \$2.77M and \$25.40M?

Solution. **No.** The distribution is strongly skewed to the right, and the rule describes unimodal, *symmetric* distributions. We should not expect it to be useful here.

What the rule would claim: about 68% of the compensations lie within one standard deviation of the mean, that is, roughly 68% of 434 $\approx$ 295 CEOs between \$2.77M and \$25.40M.

What the data actually show: 391 CEOs, about

$$\frac{391}{434} = 0.901 = 90.1\%,$$

fall in that interval — not 68%. The rule is off by more than twenty percentage points. It is not appropriate for strongly skewed data such as these.

the rule predicts ≈ 295; the truth is 391

MODELS MAKE ASSUMPTIONS

All models are wrong — but some are useful. — George Box

Whenever we model, we must say what we are assuming and then *check the condition* in the data. For the Normal model the condition is this:

Nearly Normal Condition. The shape of the data's distribution is unimodal and symmetric, and there are no obvious outliers. Check it by making a histogram.

CAREFUL

Standardizing does not change the shape of a distribution. If the data are skewed to begin with, computing z-scores will give you skewed z-scores. No arithmetic turns a skewed distribution into a Normal one.

3.7 Problem for you: boxplots

Exercise 7 (Boxplots by hand, and how an outlier looks)

Investigate how to make a boxplot, both by hand and with StatCrunch. How does an outlier appear on a boxplot?

Solution. A **boxplot** is a picture of the five-number summary, with one extra rule that decides which points get drawn separately.

By hand, in five steps.

Step 1. Compute the five-number summary: min, Q_1, median, Q_3, max.

Step 2. Compute the IQR, IQR $= Q_3 - Q_1$, and from it the two **fences**:

$$\text{lower fence} = Q_1 - 1.5\,\text{IQR}, \qquad \text{upper fence} = Q_3 + 1.5\,\text{IQR}.$$

The fences are not drawn in the final picture. They are only a criterion.

Step 3. Draw a box from Q_1 to Q_3, with a line across it at the median. The box is the middle half of the data, and its length is the IQR.

Step 4. Extend a whisker from each end of the box out to the most extreme data value that is still *inside* the fence — not to the fence itself.

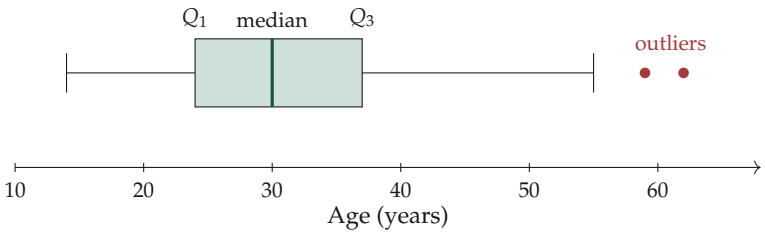
Step 5. Any value beyond a fence is plotted as an individual point. Those are the **outliers**.

So an outlier looks like a lone dot sitting out past the end of a whisker, separated from the box. That is the whole answer to the last question: the boxplot does not hide extreme values inside a whisker, it ejects them and draws them one by one.

A worked case: the *Titanic* crew ages. From the previous chapter, min $= 14$, $Q_1 = 24$, median $= 30$, $Q_3 = 37$, max $= 62$, and IQR $= 13$. The fences are

$$24 - 1.5(13) = 24 - 19.5 = 4.5, \qquad 37 + 1.5(13) = 37 + 19.5 = 56.5.$$

The minimum, 14, is above 4.5, so there are no low outliers and the lower whisker runs down to 14. But the maximum, 62, is beyond 56.5: the oldest crew members are outliers by this rule, and they are drawn as separate points, with the upper whisker stopping at the largest age that is at most 56.5.



A caution, from our own small data set. For 1, 3, 7, 22, 55 we found $Q_1 = 2$, median $= 7$, $Q_3 = 38.5$, IQR $= 36.5$. The fences are

$$2 - 1.5(36.5) = -52.75, \qquad 38.5 + 1.5(36.5) = 93.25,$$

and $55 < 93.25$, so by the rule there is *no* outlier — even though 55 looks far away from the rest. With only five values the quartiles are far apart, the IQR is large, and the fences are generous.

The 1.5 IQR rule is a convention for flagging candidates, not a verdict; in a small batch, use your eyes as well.

In StatCrunch. Enter the values in one column, then Graph → Boxplot, choose the column, and tick Use fences to identify outliers so that extreme points are drawn separately rather than swallowed by the whisker. Draw boxes horizontally matches the orientation used above. The five-number summary itself comes from Stat → Summary Stats → Columns.

> **WHY** $1.5 \times$ IQR
>
> There is nothing sacred about 1.5; it is a convention chosen so that, for data that really are Normal, only about 0.7% of values get flagged. It is wide enough not to cry wolf and narrow enough to catch genuine strays. What matters is that the criterion is *stated*, so that two people drawing the same boxplot draw the same picture.

CHAPTER 4

Scatterplots, Association and Correlation

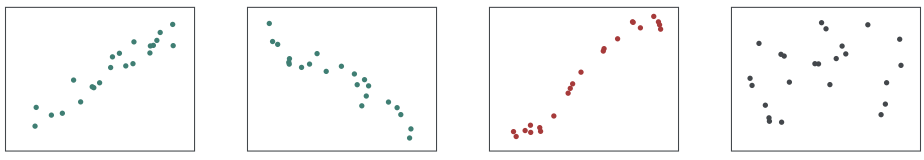
Up to here every tool has described *one* variable at a time. Now we study the relationship *between two quantitative variables*, which is where most of the interesting questions live: do students who study more get better grades, is there a connection between height and weight, do hurricane forecasts improve over the years.

4.1 Scatterplots

A **scatterplot** shows the relationship between two quantitative variables, one on each axis, with a point for each case. They are the most common display in all of statistics, and the right place to start: patterns, trends, relationships and the occasional extraordinary value are all visible at a glance. When the x-variable is time, the plot is often called a **timeplot**.

Everyone looks at scatterplots. But if asked, many people find it hard to say *what to look for* in one. There are four things, and they come in order.

> **WHAT TO LOOK FOR IN A SCATTERPLOT**
>
> **Direction.** What is my sign — positive, negative, or neither? A pattern running from the lower left to the upper right is **positive**; one running from the upper left to the lower right is **negative**.
> **Form.** Straight, curved, something exotic, or no pattern at all?
> **Strength.** How much scatter? Are the points tightly clustered in a single stream, or do they form a vague cloud through which we can barely discern any trend?
> **Unusual features.** Are there outliers or subgroups?



positive negative curved no pattern

Finally, always look for the unexpected. An **outlier** standing away from the overall pattern is almost always interesting and always deserves special attention. In the scatterplot of hurricane prediction errors against year, 1972 stands out with very high errors; it was a quiet hurricane season, but it included Hurricane Agnes, which combined with another low-pressure centre to ravage the north-eastern United States, killing 122 people. Possibly Agnes was also unusually difficult to predict.

You should also look for clusters or subgroups that stand away from the rest of the plot, or that trend in a different direction. A deviating group should raise the question of *why* it is different, and may be a clue that you should split the data instead of looking at it all together.

Exercise 1 (Comparing prices worldwide)

UBS prepared a report comparing prices, wages and other economic conditions in 73 cities around the world. Four scatterplots of pairs of those variables are described below. Describe the pattern shown by each.

A. *Food Costs* against *Cost of Women's Clothes*: a rising band of points with a few very high ones at the top.

B. *Hours to Earn an iPhone 4S* against *Average Hourly Wage*: a steep fall that flattens out into a long low tail.

C. *Clothes Index* against *Working Hours*: a wide, gently falling cloud with one or two high points.

D. *Food Costs* against *Vacation Days*: a shapeless band.

Solution. Answer the four questions — direction, form, strength, unusual features — for each one.

A. The association between *Food Costs* and *Cost of Women's Clothes* is *positive* and *straight*, *moderately strong*, with a few high outliers.

B. The association between *Hours to Earn an iPhone* and *Average Hourly Wage* is *negative* and *strong*, but the form is *not straight*: it drops steeply and then levels off. This matters, and we will come back to it.

C. The association between the *Clothes Index* and *Working Hours* is *weak*, *negative* and generally *straight*, with one or two possible high outliers.

D. There does not appear to be *any* association between *Food Costs* and *Vacation Days*.

WHICH VARIABLE GOES ON WHICH AXIS

By convention the **explanatory** variable goes on the x-axis and the **response** variable on the y-axis. The choice is yours, and it is a statement about what you want to explain: height explaining weight is a different story from weight explaining height, even though the picture is the same cloud of points. Decide which variable you are treating as the explanation *before* you draw.

4.2 Correlation

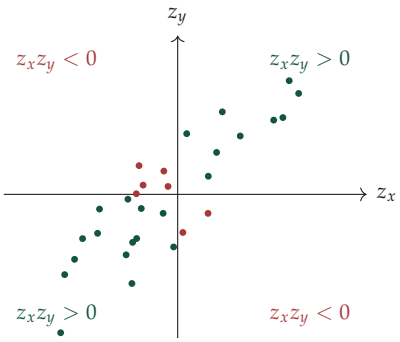
A **correlation** describes the strength and direction of a *linear* relationship between two quantitative variables. It answers questions like "do students who study more get better grades?" with a single number.

We want a measure of strength that does not depend on the units: if we had measured weight in kilograms instead of pounds and height in centimetres instead of inches, the picture would look the same and the strength should be the same number. Two steps give us exactly that.

First, subtract the means. This moves the centre of the cloud to the origin and splits the plane into four quadrants. Points in the upper right and lower left are consistent with a *positive* association; points in the other two quadrants are consistent with a *negative* one.

Second, divide by the standard deviations. This removes the units altogether. Instead of plotting (x, y) we are plotting

$$(z_x,\, z_y) = \left(\frac{x-\bar{x}}{s_x},\, \frac{y-\bar{y}}{s_y}\right).$$



For a point in a green quadrant both z-scores have the same sign, so their product $z_x z_y$ is positive; in a red quadrant the signs differ and the product is negative. Adding up those products, $\sum z_x z_y$, summarizes both the direction and the strength at once. Points far from the origin have large z-scores and contribute more. Dividing by $n-1$ keeps the sum from growing simply because there are more points, and the result is the **correlation coefficient**:

$$r = \frac{\sum z_x z_y}{n-1}.$$

Two equivalent formulas, in the original units, are convenient for computing by hand:

$$r = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\sum (x-\bar{x})^2 \sum (y-\bar{y})^2}} = \frac{\sum (x-\bar{x})(y-\bar{y})}{(n-1)\, s_x s_y}.$$

The z-score form is the one to keep in your head, because it is the one that explains what correlation *means*.

Exercise 2 (Finding the correlation coefficient by hand)

Find r for the data pairs

x	6	10	14	19	21
y	5	3	7	8	12

Solution. **Step 1: the summary statistics.**

$$\bar{x} = \frac{6+10+14+19+21}{5} = \frac{70}{5} = 14, \qquad \bar{y} = \frac{5+3+7+8+12}{5} = \frac{35}{5} = 7.$$

For the standard deviations, $\sum (x-\bar{x})^2 = 64+16+0+25+49 = 154$ and $\sum (y-\bar{y})^2 = 4+16+0+1+25 = 46$, so

$$s_x = \sqrt{\tfrac{154}{4}} = 6.20, \qquad s_y = \sqrt{\tfrac{46}{4}} = 3.39.$$

Step 2: the deviations and their products.

$x - \bar{x}$	$y - \bar{y}$	product
$6 - 14 = -8$	$5 - 7 = -2$	$(-8)(-2) = 16$
$10 - 14 = -4$	$3 - 7 = -4$	16
$14 - 14 = 0$	$7 - 7 = 0$	0
$19 - 14 = 5$	$8 - 7 = 1$	5
$21 - 14 = 7$	$12 - 7 = 5$	35
		72

Step 3: divide.

$$(n-1)\, s_x s_y = 4 \times 6.20 \times 3.39 = 84.07,$$

$$r = \frac{72}{84.07} = 0.856.$$

A strong positive linear association.

$r = 0.856$

4.3 Conditions for using a correlation

Correlation measures the strength of *linear* association between two *quantitative* variables. To interpret one we must assume there is a true underlying linear relationship. We cannot know that, but we can check whether it is plausible. What we assume about the variables we call assumptions; what we can check by looking at the data we call **conditions**.

THREE CONDITIONS, AND ALL THREE ARE QUICK

Quantitative Variables Condition. Correlation is only about quantitative variables. Do not call an association involving a categorical variable a correlation.
Straight Enough Condition. Look at the scatterplot and ask whether it looks reasonably straight. A judgment call, but not a hard one.
No Outliers Condition. Outliers distort a correlation dramatically — they can make a weak association look strong, a strong one look weak, and they can even flip the sign. Outliers are easy to see in a scatterplot, so to check this, just look.

Exercise 3 (Correlations for the four city plots)

The correlations for plots A to D of the previous exercise are 0.614, -0.791, -0.388 and -0.040. Check the conditions, and interpret the correlations you are entitled to interpret.

Solution. All of the variables examined are quantitative, and none of the plots shows an outlier. But plot B, *Hours to Earn an iPhone* against *Average Wage*, is *not straight*, so it fails the Straight Enough Condition and **its correlation of** -0.791 **is not an appropriate summary** — even though the association there is obviously strong. This is the most important line in the exercise.

For the other three:

$r = 0.614$ between *Food Costs* and *Women's Clothing Costs* indicates a moderately strong positive association.

$r = -0.388$ between the *Clothes Index* and *Working Hours* indicates a moderately weak negative association.

$r = -0.040$ between *Food Costs* and *Vacation Days* is so small that it suggests there may be no linear association between them at all.

B is disqualified: strong, but not linear

CAREFUL

A correlation near zero does not mean "no relationship". It means "no *linear* relationship". Plot B has a very strong relationship and a correlation that we are not allowed to use. Always look at the picture before you trust the number.

4.4 Properties of the correlation coefficient

WHAT r DOES AND DOES NOT DO

- The **sign** of r gives the direction of the association.
- r is always between -1 and $+1$. It *can* equal ± 1, but only if every point falls exactly on a single straight line, which is unusual in real data.
- Correlation treats x and y **symmetrically**: the correlation of x with y is the same as that of y with x.
- Correlation has **no units**, which is why it is useful when the units are vague to begin with (an IQ score, a personality index). Do not report it as a percentage: a percentage is a percentage *of something*, and r has no something.
- Correlation is **not affected by changes in centre or scale** of either variable. It depends only on the z-scores.
- Correlation measures the strength of the **linear** association only. Variables can be strongly associated and still have a small r.
- Correlation is **sensitive to outliers**. A single outlying value can make a small correlation large or a large one small.

CHAPTER 5

Linear Regression

The correlation tells us that the linear association between two variables is fairly strong. It does not tell us *what the line is*. That is what regression does.

5.1 The linear model

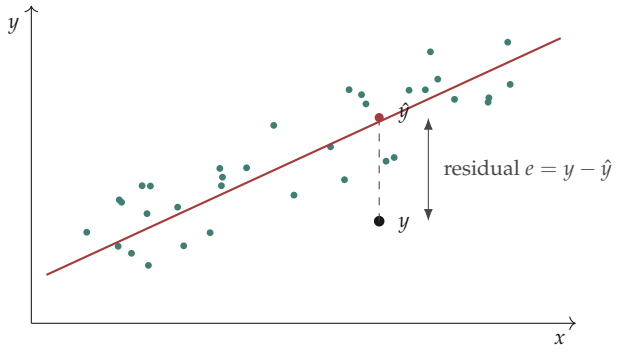
From algebra, a straight line is $y = mx + b$. In statistics we use the same form with different notation:

$$\hat{y} = b_0 + b_1 x.$$

We write $\hat{y}$ — "y-hat" — to emphasize that the points satisfying this equation are our **predicted** values, not the actual data values, which scatter around the line. The numbers b_0 and b_1 are the **coefficients** of the model: b_1 is the **slope**, telling how rapidly $\hat{y}$ changes with respect to x, and b_0 is the **intercept**, telling where the line crosses the y-axis.

No line can pass through all the points, and the best line might not pass through *any* of them. The estimate made from the model is the predicted value $\hat{y}$; the difference between what we observed and what the model predicted is the **residual**:

$$\text{Residual} = \text{Observed value} - \text{Predicted value}, \qquad e = y - \hat{y}.$$



THE SIGN OF A RESIDUAL

A point that lies **above** the line has $y > \hat{y}$, so its residual is **positive**. A point **below** the line has $y < \hat{y}$, so its residual is **negative**.
The mnemonic is the definition itself: observed minus predicted. If the data point is higher than the line predicted, the subtraction comes out positive.

5.2 Finding the least squares line

Least squares lines are commonly called **regression lines**; for now, "regression" simply means "the linear model fit by least squares". Two formulas give it, and both come from things we

already have:

$$b_1 = r\,\frac{s_y}{s_x}, \qquad b_0 = \bar{y} - b_1\bar{x}.$$

The second one deserves a word. The point $(\bar{x}, \bar{y})$ — the mean of x paired with the mean of y — *always lies on the regression line*. So substituting it into $\hat{y} = b_0 + b_1 x$ gives $\bar{y} = b_0 + b_1\bar{x}$, and solving for b_0 gives the formula above. The line is pinned at the centre of the data and tilted by the correlation.

> **Exercise 1** (The Burger King regression line)
>
> For 122 items on the Burger King menu, *Protein* (g) and *Fat* (g) have the summary statistics
>
> $$\bar{x} = 18.0 \text{ g}, \quad s_x = 13.5 \text{ g}, \quad \bar{y} = 24.8 \text{ g}, \quad s_y = 16.2 \text{ g}, \quad r = 0.76.$$
>
> Find the regression line for predicting fat from protein, and interpret both coefficients.

Solution. The slope.

$$b_1 = r\,\frac{s_y}{s_x} = 0.76 \times \frac{16.2 \text{ g fat}}{13.5 \text{ g protein}} = 0.91\ \frac{\text{g fat}}{\text{g protein}}.$$

The intercept.

$$b_0 = \bar{y} - b_1\bar{x} = 24.8 \text{ g fat} - 0.91\ \frac{\text{g fat}}{\text{g protein}} \times 18.0 \text{ g protein} = 8.4 \text{ g fat}.$$

The model.

$$\widehat{Fat} = 8.4 + 0.91\,Protein.$$

Interpreting the slope. A Burger King item with one more gram of protein can be expected, on average, to have 0.91 more grams of fat. Less formally: BK foods pack about 0.91 grams of fat per gram of protein. Slopes are always expressed in y-units per x-unit — whenever you hear a phrase like "students per teacher" or "kilobytes per second", think slope.

Interpreting the intercept. Algebraically, 8.4 g is what the line gives when protein is zero: an item with no protein would have, on average, about 8.4 grams of fat. Is that reasonable? There are two items on the menu with 0 grams of protein (apple fries and applesauce) and neither has any fat, so the model is not being literal here. Often 0 is not even a plausible value for x — the year 0, a baby weighing 0 grams — and then the intercept serves only as a starting value for predictions and should not be interpreted as a meaningful predicted value.

$\widehat{Fat} = 8.4 + 0.91\,Protein$

> **Exercise 2** (Predicting, and the residual)
>
> The BK Tendercrisp chicken sandwich has 31 grams of protein and actually contains 22 grams of fat. What does the model predict, and what is the residual?

Solution. Predict. Substitute $x = 31$:

$$\widehat{Fat} = 8.4 + 0.91(31) = 8.4 + 28.21 = 36.6 \text{ g}.$$

Carrying full precision rather than the rounded coefficients, the predicted value to one decimal place is 36.7 g. *Always keep full precision until you report the result* — rounding the slope to two decimals and then multiplying by 31 has already cost us a tenth of a gram.

The residual.

$$e = y - \hat{y} = 22 - 36.7 = -14.7 \text{ g of fat.}$$

It is negative, which says the sandwich has *less* fat than the model predicted from its protein: the point lies below the line.

$\hat{y} = 36.7$ g, $\quad e = -14.7$ g

5.3 Conditions for regression

To use a regression model we check the same three conditions as for correlation, for the same reasons:

THE THREE CONDITIONS AGAIN

Quantitative Variables Condition — both variables measured, with units.
Straight Enough Condition — the scatterplot looks reasonably straight.
No Outliers Condition — no point stands away from the pattern.
If you made the scatterplot first, as you should, checking all three costs no extra effort.

5.4 Examining the residuals

Residuals tell us whether the model makes sense. When a regression model is appropriate it should capture the underlying relationship, and nothing interesting should be left behind. So after fitting a model we plot the residuals, hoping to find . . . nothing interesting.

THE MOST BORING SCATTERPLOT YOU HAVE EVER SEEN

A plot of the residuals against x should have no direction, no shape and no bends; it should stretch out horizontally with about the same amount of scatter throughout, and it should have no outliers.
If you see any of these features, investigate: the pattern you can see in the residuals is precisely what the model missed.

Exercise 3 (Hurricane Katrina)

The central barometric pressure of a hurricane predicts its maximum wind speed; the relationship is straight, strong and negative, with $r = -0.898$. Fitting the line gives

$$\widehat{MaxWindSpeed} = 1031.24 - 0.9748\, CentralPressure.$$

Hurricane Katrina had a central pressure of 920 millibars. What does the model predict for her maximum wind speed, and how good is that prediction, given that Katrina's actual wind speed was measured at 150 knots?

Solution. **Predict.** Substitute $x = 920$:

$$\widehat{MaxWindSpeed} = 1031.24 - 0.9748(920) = 1031.24 - 896.82 = 134.4 \text{ knots}.$$

The residual.

$$e = y - \hat{y} = 150 - 134.4 = 15.6 \text{ knots}.$$

Interpret. The residual is positive and sizeable: Katrina's winds were about 15.6 knots *stronger* than the model predicted from her pressure alone. On a scale where the predictions run from roughly 75 to 140 knots, an error of 15.6 knots is not small — the model gets the general relationship right but should not be trusted to the knot for an individual storm.

Note also the sign of the slope, -0.9748 knots per millibar: lower pressure means stronger winds, which is why the central pressure is used as a measure of a hurricane's strength in the first place.

$\hat{y} = 134.4$ kt, $\quad e = +15.6$ kt

> **CAREFUL**
>
> Do not extrapolate. The Burger King model was built from items with between 0 and about 70 grams of protein; asking it about an item with 300 grams is asking the line a question the data never answered. A regression line is a summary of the range it was fitted over, not a law of nature that extends forever.

5.5 Where we are

It is worth stopping to see how far the tools now reach. We take a small group drawn from a population, and we can give a fairly accurate description of it: shape, centre, spread. We can find linear relationships between two quantitative variables, measure their strength with a correlation, and — once the conditions are checked — model the association with a regression line and make predictions with it.

What we have *not* yet worried about is the step that comes before all of it: how that small group was chosen. Everything above assumes the data arrived in a trustworthy way. That is the problem of randomness, and it is where the course goes next.

CHAPTER 6

From Randomness to Probability

Everything so far assumed that the small group we studied was chosen *randomly* from the population. It is time to look at that assumption itself, and the language we need for it is probability.

6.1 Random phenomena

For us, a **random phenomenon** is a situation in which we know what outcomes can possibly occur, but we do not know which particular one will happen.

Each occasion on which we observe a random phenomenon is called a **trial**. At each trial we note the value of the phenomenon, and call that the trial's **outcome**. The collection of all possible outcomes is the **sample space**, written S.

6.2 Modelling probability

It is easy to find probabilities for events made up of several *equally likely* outcomes: count the outcomes the event contains, and divide by the total number of possible outcomes,

$$P(\mathbf{A}) = \frac{\#\text{ outcomes in } \mathbf{A}}{\#\text{ of possible outcomes}}.$$

Exercise 1 (A face card)

What is the probability of drawing a face card (jack, queen or king) from a standard deck?

Solution. A deck has 52 cards, all equally likely to be drawn. Face cards are the jack, queen and king in each of the four suits, so there are $3 \times 4 = 12$ of them:

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13} \approx 0.231.$$

$P(\text{face card}) = \frac{3}{13}$

CAREFUL

"Straightforward" is not the same as "easy". Flip two coins and the sample space is $S = \{HH, HT, TH, TT\}$, so $P(\text{exactly one head}) = 2/4 = 1/2$. Flip 100 coins and the sample space has more than 1.2×10^{30} outcomes — and that is only the denominator. Counting is a subject of its own.

Personal probability

What is the probability that your grade in this course will be an A? You can probably name a number, and it should be between 0 and 1 — but it is not an empirical probability, because you cannot take the course over and over, forgetting everything each time.

We use probability language in everyday speech to express a degree of uncertainty without basing it on long-run relative frequencies or on a mathematical model. That third kind is a **subjective**, or personal, probability. Personal probabilities may be based on experience, but they do not display the consistency we are going to need, so in this course we stick to formally defined probabilities. Be alert to the difference.

THE FIRST THREE RULES FOR WORKING WITH PROBABILITY

1. Make a picture.
2. Make a picture.
3. Make a picture.

We are dealing with probabilities now rather than data, and the rules do not change. The usual picture is a **Venn diagram**. Experienced statisticians draw them too.

6.3 Formal probability

For some people "50/50" means something vague like "I don't know". When we discuss probabilities it takes the precise meaning of *equally likely*, so we need formal rules.

THE FIVE RULES

Rule 1 — Probability is a number between 0 and 1. For any event $\mathbf{A}$, $0 \leq P(\mathbf{A}) \leq 1$. If the probability is 0 the event never occurs; if it is 1 it always occurs.
Rule 2 — Probability Assignment Rule. The set of all possible outcomes of a trial must have probability 1: $P(S) = 1$.
Rule 3 — Complement Rule. The probability of an event *not* occurring is 1 minus the probability that it does: $P(\mathbf{A}^C) = 1 - P(\mathbf{A})$.
Rule 4 — Addition Rule. For two *disjoint* events, $P(\mathbf{A} \text{ or } \mathbf{B}) = P(\mathbf{A}) + P(\mathbf{B})$. Disjoint (or mutually exclusive) events have no outcomes in common.
Rule 5 — Multiplication Rule. For two *independent* events, $P(\mathbf{A} \text{ and } \mathbf{B}) = P(\mathbf{A}) \times P(\mathbf{B})$. Independence means the outcome of one does not influence the other.

Exercise 2 (The light at College and Main)

When you arrive at the corner of College and Main the light is green about 35% of the time, so $P(\text{green}) = 0.35$. The light is red, green or yellow, and $P(\text{yellow}) = 0.04$.

(a) What is the probability the light is *not* green?

(b) What is the probability the light is red?

Solution. **(a)** "Not green" is the complement of "green", so by Rule 3

$$P(\text{not green}) = 1 - P(\text{green}) = 1 - 0.35 = 0.65.$$

There is a 65% chance you will not have a green light.

(b) The light must be red, green or yellow. Green and yellow are disjoint — the light cannot be both at once — so Rule 4 applies:

$$P(\text{green or yellow}) = 0.35 + 0.04 = 0.39.$$

Red is the only remaining alternative, and the three probabilities must add to 1, so by Rule 3 again

$$P(\text{red}) = 1 - P(\text{green or yellow}) = 1 - 0.39 = 0.61.$$

(a) 0.65 (b) 0.61

Exercise 3 (A week of red lights)

With $P(\text{red}) = 0.61$:

(a) What is the probability the light is red both Monday and Tuesday?

(b) What is the probability you do not meet a red light until Wednesday?

(c) What is the probability you have to stop *at least once* during the five weekdays?

Solution. The colour seen on one morning does not influence the colour seen on another, so these are independent events and Rule 5 applies.

(a)

$$P(\text{red Mon and red Tue}) = (0.61)(0.61) = 0.3721,$$

about a 37% chance.

(b) "No red until Wednesday" means not red on Monday, not red on Tuesday, and red on Wednesday. First, $P(\text{not red}) = 1 - 0.61 = 0.39$. Then

$$(0.39)(0.39)(0.61) = 0.092781,$$

about a 9% chance.

(c) "At least once" means stopping 1, 2, 3, 4 or 5 times — five separate calculations. Far easier to use the complement: the opposite of stopping at least once is *never* stopping.

$$P(\text{at least one red in 5 days}) = 1 - P(\text{no red in 5 days}) = 1 - (0.39)^5 = 1 - 0.0090 = 0.991.$$

Over a 99% chance of hitting at least one red light that week.

(a) 0.372 (b) 0.093 (c) 0.991

"AT LEAST ONCE"

Whenever a question asks for the probability of *at least one* success, look at the complement first: "at least one" is the opposite of "none", and "none" is a single product. This trick will keep working for the rest of the course.

Exercise 4 (Four flights in a row)

Your job requires you to fly Atlanta to Houston every Monday. The airline reports the flight is on time 85% of the time. What is the chance it is on time four Mondays in a row?

Solution. Whether the flight is on time one week does not influence the next, so assuming independence,

$$(0.85)^4 = 0.85 \times 0.85 \times 0.85 \times 0.85 = 0.522,$$

just over a coin flip.

0.522

CAREFUL

Many statistical methods need an **Independence Assumption** — and assuming independence does not make it true. Before using the Multiplication Rule, think about whether the assumption is reasonable. Four Mondays in a row during hurricane season are not independent at all.

6.4 The General Addition Rule

Rule 4 required the events to be disjoint. When they overlap, adding the two probabilities counts the overlap twice, so we subtract it back:

$$P(\mathbf{A} \text{ or } \mathbf{B}) = P(\mathbf{A}) + P(\mathbf{B}) - P(\mathbf{A} \text{ and } \mathbf{B}).$$

This is the same addition rule we met for overlapping areas in Chapter 1, word for word.

Exercise 5 (Relationships and sports)

A survey of college students found that 33% were currently in a relationship, 25% were involved in intercollegiate or club sports, and 11% said yes to both. What is the probability that a randomly selected student either is in a relationship or is involved in sports?

Solution. Let **R** be "in a relationship" and **S** "involved in sports". These are not disjoint — 11% are both — so we need the general rule:

$$P(\mathbf{R} \text{ or } \mathbf{S}) = P(\mathbf{R}) + P(\mathbf{S}) - P(\mathbf{R} \text{ and } \mathbf{S}) = 0.33 + 0.25 - 0.11 = 0.47.$$

0.47

CAREFUL

This use of "or" includes "or both", which is its usual meaning in English and always its meaning in probability. To exclude the students who are in both, you have to ask a different question — see part (c) of the next exercise.

Exercise 6 (The same survey, with a Venn diagram)

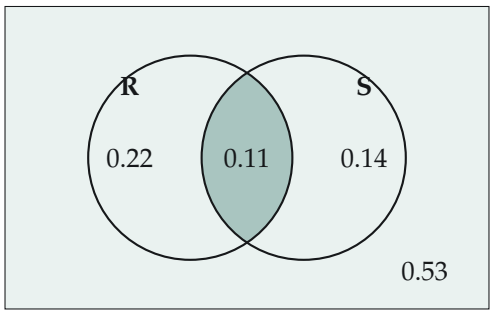
With $P(\mathbf{R}) = 0.33$, $P(\mathbf{S}) = 0.25$ and $P(\mathbf{R} \text{ and } \mathbf{S}) = 0.11$, find the probability that a randomly selected student

(a) is not in a relationship and is not involved in sports;

(b) is in a relationship but not involved in sports;

(c) either is in a relationship or is involved in sports, *but not both*.

Solution. Build the picture first. The overlap is 0.11. The part of **R** outside the overlap is $0.33 - 0.11 = 0.22$, and the part of **S** outside it is $0.25 - 0.11 = 0.14$. Those three regions total $0.22 + 0.11 + 0.14 = 0.47$, so the region outside both circles is $1 - 0.47 = 0.53$.



(a) Neither: that is the region outside both circles,

$$P(\mathbf{R}^C \text{ and } \mathbf{S}^C) = 0.53.$$

(b) In a relationship but not in sports: the left crescent,

$$P(\mathbf{R} \text{ and } \mathbf{S}^C) = 0.33 - 0.11 = 0.22.$$

(c) One or the other but not both: the two crescents together,

$$0.22 + 0.14 = 0.36.$$

Equivalently, take the union and remove the overlap *twice* — once because it should not be there, and once more because the union already subtracted it:

$$P(\mathbf{R}) + P(\mathbf{S}) - 2P(\mathbf{R} \text{ and } \mathbf{S}) = 0.33 + 0.25 - 2(0.11) = 0.36.$$

(a) 0.53 (b) 0.22 (c) 0.36

6.5 Conditional probability

Two psychologists surveyed 478 children in grades 4, 5 and 6 in Michigan, asking whether their primary goal was to get good grades, to be popular, or to be good at sports.

	Grades	Popular	Sports	Total
Boy	117	50	60	227
Girl	130	91	30	251
Total	247	141	90	478

Taking the sample space to be these 478 students, each equally likely to be selected,

$$P(\text{girl}) = \frac{251}{478} = 0.525, \qquad P(\text{girl and popular}) = \frac{91}{478} = 0.190, \qquad P(\text{sports}) = \frac{90}{478} = 0.188.$$

Now suppose we are told the selected student is a girl. Does that change the probability that the goal is sports? It does. We restrict our attention to the girls' row: of the 251 girls, only 30 chose sports, so

$$P(\text{sports} \mid \text{girl}) = \frac{30}{251} = 0.120.$$

For boys, of 227, 60 chose sports, so $P(\text{sports} \mid \text{boy}) = 60/227 = 0.264$ — more than twice as likely.

CONDITIONAL PROBABILITY

$$P(\mathbf{B} \mid \mathbf{A}) = \frac{P(\mathbf{A} \text{ and } \mathbf{B})}{P(\mathbf{A})}, \qquad \text{provided } P(\mathbf{A}) \neq 0.$$

Read it as "the probability of **B** given **A**". To find it, restrict attention to the outcomes in **A**, and ask in what fraction of *those* **B** also occurred.
The restriction $P(\mathbf{A}) \neq 0$ is not a technicality: we cannot be "given" that **A** happened if **A** never happens.

Counts or probabilities give the same answer, because the total cancels:

$$\frac{P(\text{sports and girl})}{P(\text{girl})} = \frac{30/478}{251/478} = \frac{30}{251} = 0.12.$$

Exercise 7 (Reversing the condition)

Using the same table, what is the probability that the student is a girl, given that the goal is popularity?

Solution. Now the condition is the *column*, not the row:

$$P(\text{girl} \mid \text{popular}) = \frac{P(\text{girl and popular})}{P(\text{popular})} = \frac{91/478}{141/478} = \frac{91}{141} = 0.65.$$

Compare this with $P(\text{popular} \mid \text{girl}) = 91/251 = 0.36$. The two are completely different questions with completely different answers, even though both use the 91.

0.65

CAREFUL

$P(\mathbf{B} \mid \mathbf{A})$ and $P(\mathbf{A} \mid \mathbf{B})$ are not the same number, and confusing them is one of the most consequential errors in applied statistics. It is the difference between "what fraction of the sick test positive" and "what fraction of those who test positive are sick".

Exercise 8 (Meeting an athlete)

In the college survey, 33% are in a relationship, 25% do sports, 11% both. You meet a student who is on a sports team. What is the probability your new acquaintance is in a relationship?

Solution. We are given that the student does sports, so we condition on $\mathbf{S}$:

$$P(\mathbf{R} \mid \mathbf{S}) = \frac{P(\mathbf{R} \text{ and } \mathbf{S})}{P(\mathbf{S})} = \frac{0.11}{0.25} = 0.44.$$

Notice that 0.44 is higher than $P(\mathbf{R}) = 0.33$: on this campus, knowing that a student plays sports makes it *more* likely they are in a relationship.

0.44

6.6 The General Multiplication Rule

Rule 5 required independence. Rearranging the definition of conditional probability removes that requirement:

$$P(\mathbf{A} \text{ and } \mathbf{B}) = P(\mathbf{A}) \times P(\mathbf{B} \mid \mathbf{A}).$$

In words: the probability that both occur is the probability that $\mathbf{A}$ occurs, times the probability that $\mathbf{B}$ *also* occurs given that $\mathbf{A}$ has. There is nothing special about which event we call $\mathbf{A}$, so equally

$$P(\mathbf{A} \text{ and } \mathbf{B}) = P(\mathbf{B}) \times P(\mathbf{A} \mid \mathbf{B}).$$

6.7 Independence

Now we can say formally what independence means. Events $\mathbf{A}$ and $\mathbf{B}$ are **independent** if — and only if —

$$P(\mathbf{B} \mid \mathbf{A}) = P(\mathbf{B}).$$

That is, knowing that $\mathbf{A}$ happened does not change the probability of $\mathbf{B}$.

The simple Multiplication Rule is just the special case: the general rule says $P(\mathbf{A} \text{ and } \mathbf{B}) = P(\mathbf{A}) \times P(\mathbf{B} \mid \mathbf{A})$ always, and when the events are independent we may write $P(\mathbf{B})$ in place of $P(\mathbf{B} \mid \mathbf{A})$, recovering $P(\mathbf{A} \text{ and } \mathbf{B}) = P(\mathbf{A}) \times P(\mathbf{B})$.

Exercise 9 (Checking independence in the table)

Is the goal of good grades independent of the student's sex? Is the goal of sports?

Solution. **Grades.** Compare the conditional probability with the unconditional one:

$$P(\text{grades} \mid \text{girl}) = \frac{130}{251} = 0.52, \qquad P(\text{grades}) = \frac{247}{478} = 0.52.$$

To two decimal places these agree, so choosing good grades as a goal can be considered independent of sex.

Sports.

$$P(\text{sports}) = \frac{90}{478} = 0.188 = 18.8\%, \qquad P(\text{sports} \mid \text{boy}) = \frac{60}{227} = 0.264 = 26.4\%.$$

These are not equal, so choosing success in sports as a goal is *not* independent of the student's sex.

grades: independent sports: not independent

Exercise 10 (Independent or disjoint?)

In the college survey, are being in a relationship and being involved in sports independent? Are they disjoint?

Solution. **Independent?** No. We computed $P(\mathbf{R} \mid \mathbf{S}) = 0.44$, while $P(\mathbf{R}) = 0.33$. Knowing a student does sports changes the probability, so the events are not independent: on this campus students involved in sports are more likely to be in a relationship.

Disjoint? No. Disjoint would mean they never happen together, and in fact 11% of students do both.

neither independent nor disjoint

INDEPENDENT $\neq$ DISJOINT

Both words suggest separation, and they are opposites in a precise sense: *disjoint events can never be independent.*
Take the two disjoint events "you get an A in this course" and "you get a B in this course". Suppose you learn that you *did* get an A. What is now the probability that you got a B? Zero — you cannot get both. Knowing the first event occurred changed the probability of the second, from whatever it was down to 0. That is precisely a failure of independence. So disjoint events have no outcomes in common, which means that if one occurs the other cannot, which is the strongest possible dependence. A common error is to treat disjoint events as independent and apply the simple Multiplication Rule. Do not make that mistake.

CHAPTER 7

Random Variables

An insurance company offers a "death and disability" policy that pays \$10,000 if you die and \$5000 if you are permanently disabled, and charges a premium of \$50 a year. Is the company likely to make a profit? To answer that we need a new object.

7.1 Random variables and probability models

The amount the company pays out on an individual policy is a **random variable**, because its numeric value is based on the outcome of a random event. We write a random variable with a capital letter, usually X, and a particular value it can take with the corresponding lowercase letter x.

Here x can be \$10,000, \$5000 or \$0. Because we can *list* all the outcomes, X is a **discrete** random variable; otherwise we would call it **continuous**. The collection of all possible values together with their probabilities is the **probability model** of the variable.

Suppose the death rate in any year is 1 in 1000, and another 2 in 1000 suffer some kind of disability. Then the model is:

Policyholder outcome	Payout x	$P(X = x)$
Death	10,000	1/1000
Disability	5000	2/1000
Neither	0	997/1000

7.2 Centre: the expected value

EXPECTED VALUE

$$\mu = E(X) = \sum x\, P(x).$$

Multiply each possible value by the probability that it occurs, and add.
The expected value is a *parameter* of the model, not a summary of data, which is why it gets a Greek letter. We do not estimate it; we assume the probabilities are known and calculate it.

Exercise 1 (The insurance policy)

What is the expected payout per policy? Is the company likely to profit on a \$50 premium?

Solution. It helps to imagine exactly 1000 clients, of whom 1 dies and 2 are disabled — just what the probabilities say. The total payout would be

$$\frac{10{,}000(1) + 5000(2) + 0(997)}{1000} = \frac{20{,}000}{1000} = \$20.$$

Written with the probabilities instead,

$$\mu = E(X) = \$10{,}000\left(\frac{1}{1000}\right) + \$5000\left(\frac{2}{1000}\right) + \$0\left(\frac{997}{1000}\right) = \$20.$$

Charging \$50 and paying out \$20 on average, the company expects a profit of \$30 per policy.

$E(X) = \$20$

Exercise 2 (The Lucky Lovers special)

On Valentine's Day the *Quiet Nook* brings the four aces to the table, shuffled and face down. The couple turns one over. A black ace means they owe the full amount; the ace of hearts earns a \$20 discount; the ace of diamonds lets them turn over one of the remaining three cards, earning \$10 if that one is the ace of hearts. What is the expected discount?

Solution. **Build the model first.** Let X be the discount.

The \$20 discount happens exactly when the first card is the ace of hearts:

$$P(X = 20) = P(A\heartsuit) = \tfrac{1}{4}.$$

The \$10 discount needs *two* things: the ace of diamonds first, and then the ace of hearts from the three that remain. That is a general multiplication:

$$P(X = 10) = P(A\diamondsuit) \times P(A\heartsuit \mid A\diamondsuit) = \tfrac{1}{4} \times \tfrac{1}{3} = \tfrac{1}{12}.$$

Everything else gives nothing, and rather than enumerate those cases we take the complement:

$$P(X = 0) = 1 - \left(\tfrac{1}{4} + \tfrac{1}{12}\right) = 1 - \tfrac{4}{12} = \tfrac{2}{3}.$$

Outcome	$A\heartsuit$	$A\diamondsuit$, then $A\heartsuit$	Black ace
x	20	10	0
$P(X = x)$	$\frac{1}{4}$	$\frac{1}{12}$	$\frac{2}{3}$

Then compute.

$$E(X) = 20 \cdot \tfrac{1}{4} + 10 \cdot \tfrac{1}{12} + 0 \cdot \tfrac{2}{3} = 5 + \tfrac{10}{12} = \tfrac{70}{12} \approx \$5.83.$$

$E(X) \approx \$5.83$

CAREFUL

Always check that your probabilities add to 1 before computing an expected value:

$\frac{1}{4} + \frac{1}{12} + \frac{2}{3} = \frac{3+1+8}{12} = 1$. If they do not, the model is wrong and the expected value is meaningless.

7.3 Spread: the standard deviation

The expected value is not what happens to any *particular* policyholder. No individual policy costs the company \$20: some receive big payouts, most nothing. The company must anticipate that variability.

VARIANCE AND STANDARD DEVIATION OF A RANDOM VARIABLE

$$\sigma^2 = \mathrm{Var}(X) = \sum (x-\mu)^2 P(x), \qquad \sigma = SD(X) = \sqrt{\mathrm{Var}(X)}.$$

Same idea as for data: take each deviation from the mean, square it, and average — except that here "average" means weighting by the probabilities rather than dividing by $n-1$.

Exercise 3 (Spread of the insurance payout)

Find the standard deviation of the payout X.

Solution. The mean is $\mu = \$20$. Build the deviations, square them, weight them.

Outcome	x	$P(X=x)$	$x-\mu$
Death	10,000	1/1000	9980
Disability	5000	2/1000	4980
Neither	0	997/1000	−20

$$\mathrm{Var}(X) = 9980^2\left(\tfrac{1}{1000}\right) + 4980^2\left(\tfrac{2}{1000}\right) + (-20)^2\left(\tfrac{997}{1000}\right) = 149{,}600,$$

and so

$$SD(X) = \sqrt{149{,}600} \approx \$386.78.$$

An average payout of \$20 with a standard deviation of \$387: the *average* is small, but the *variability* is enormous, which is exactly why insurance companies need many policyholders.

$SD(X) \approx \$386.78$

Exercise 4 (Spread of the Lucky Lovers discount)

What is the standard deviation of the discounts?

Solution. With $\mu = 5.83$,

$$\mathrm{Var}(X) = (20-5.83)^2 \cdot \tfrac{1}{4} + (10-5.83)^2 \cdot \tfrac{1}{12} + (0-5.83)^2 \cdot \tfrac{2}{3}$$

$$= 200.79 \cdot \tfrac{1}{4} + 17.39 \cdot \tfrac{1}{12} + 33.99 \cdot \tfrac{2}{3} = 50.20 + 1.45 + 22.66 \approx 74.31,$$

$$SD(X) = \sqrt{74.31} \approx \$8.62.$$

$\mu = \$5.83, \qquad \sigma \approx \$8.62$

7.4 Shifting and combining random variables

Adding a constant

The company pays out \$20 on average and charges \$50, so it profits \$30 per policy. Lower the premium by \$5 and the expected profit drops by \$5. But the *variability* does not change at all: shifting every value by the same amount moves the centre and leaves the spread alone, exactly as it did for data.

$$E(X \pm c) = E(X) \pm c, \qquad \mathrm{Var}(X \pm c) = \mathrm{Var}(X).$$

Exercise 5 (Adding a coupon)

Couples at the *Quiet Nook* expect discounts averaging \$5.83 with a standard deviation of \$8.62. The restaurant also hands out coupons worth \$5 off any one meal, one per table. If every couple brings a coupon, what are the mean and standard deviation of their total discount?

Solution. Let $D = X + 5$.

$$E(D) = E(X) + 5 = 5.83 + 5 = \$10.83,$$
$$\mathrm{Var}(D) = \mathrm{Var}(X) = 8.62^2, \qquad SD(D) = \$8.62.$$

Every couple gains exactly \$5, so nothing about the *variation* between couples changes.

$\mu = \$10.83, \qquad \sigma = \$8.62$

Multiplying by a constant

Multiplying every value by a constant multiplies the mean by that constant and the variance by its *square*:

$$E(aX) = a\,E(X), \qquad \mathrm{Var}(aX) = a^2\,\mathrm{Var}(X).$$

Exercise 6 (Double the love)

When two couples dine together on a single check, the restaurant doubles the offer: \$40 for the ace of hearts on the first card and \$20 on the second. What are the mean and standard deviation of the discounts for such foursomes?

Solution. Every discount is doubled, so the variable is $2X$.

$$E(2X) = 2E(X) = 2(5.83) = \$11.66,$$
$$\mathrm{Var}(2X) = 2^2\,\mathrm{Var}(X) = 4(8.62^2) = 297.22, \qquad SD(2X) = \sqrt{297.22} = \$17.24.$$

$\mu = \$11.66, \qquad \sigma = \$17.24$

Adding two random variables

SUMS AND DIFFERENCES

$$E(X \pm Y) = E(X) \pm E(Y)$$

always. And *if X and Y are independent*,

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y).$$

Read that second line twice. The variances **add** even for a *difference*. Variability never cancels; combining two uncertain things produces something more uncertain, not less.

Exercise 7 (One check or two?)

You and your partner go to the *Quiet Nook* with another couple and agree to share any benefit. Does it matter whether you pay separately or together?

Solution. Paying together gives the doubled discount $2X$, computed above: mean \$11.66, standard deviation \$17.24.

Paying separately gives two independent discounts X_1 and X_2 — the cards are reshuffled for each couple's turn — and the total is $T = X_1 + X_2$:

$$E(T) = E(X_1) + E(X_2) = 5.83 + 5.83 = \$11.66,$$

$$\text{Var}(T) = \text{Var}(X_1) + \text{Var}(X_2) = 8.62^2 + 8.62^2 = 148.61, \qquad SD(T) = \sqrt{148.61} = \$12.19.$$

So: the expected saving is the same either way, \$11.66. But separate checks give *less variation* in the total, \$12.19 against \$17.24. If you dislike surprises, pay separately.

same mean; σ = \$12.19 separately vs \$17.24 together

CAREFUL

$2X$ and $X_1 + X_2$ have the same mean and *different* standard deviations, and this is the single most important distinction in the chapter. $2X$ is one draw, doubled: whatever happens, it happens twice over. $X_1 + X_2$ is two separate draws, and a bad one can be offset by a good one. In symbols,

$$SD(2X) = 2\sigma = 17.24 \qquad \text{but} \qquad SD(X_1 + X_2) = \sigma\sqrt{2} = 12.19.$$

Never write $\text{Var}(X_1 + X_2) = \text{Var}(2X)$.

Exercise 8 (Working with differences)

Up the street, the *Wise Fool* has a competing promotion whose discounts average \$10.00 with a standard deviation of \$15.00. How much more can you expect to save at the *Wise Fool*, and with what standard deviation?

Solution. Let W be the *Wise Fool* discount and X the *Quiet Nook* discount. They are different promotions at separate restaurants, so they are independent.

$$E(W - X) = E(W) - E(X) = 10.00 - 5.83 = \$4.17,$$

$$SD(W - X) = \sqrt{\text{Var}(W) + \text{Var}(X)} = \sqrt{15^2 + 8.62^2} = \sqrt{299.30} \approx \$17.30.$$

Note again that we *added* the variances even though we subtracted the means.

$\mu = \$4.17, \qquad \sigma \approx \$17.30$

> **Exercise 9** (Forty couples)
>
> The owner expects to serve 40 couples on Valentine's Day. What is the expected total of the discounts given, and with what standard deviation?

Solution. Let $T = X_1 + X_2 + \cdots + X_{40}$. Means always add:

$$E(T) = 40 \times 5.83 = \$233.20.$$

Reshuffling the cards between couples makes the discounts independent, so the variances add too:

$$\text{Var}(T) = 40 \times 8.62^2, \qquad SD(T) = 8.62\sqrt{40} \approx \$54.52.$$

$\mu = \$233.20, \qquad \sigma \approx \$54.52$

> **THE $\sqrt{n}$ THAT RUNS THROUGH THE REST OF THE COURSE**
>
> Forty couples cost the owner 40 times the average, but the uncertainty grows only by $\sqrt{40} \approx 6.3$. Totals grow like n; their standard deviations grow like $\sqrt{n}$. That mismatch is why large samples are more reliable than small ones, and it is the engine of every inference method that follows.

CHAPTER 8

Probability Models

A cereal manufacturer puts pictures of famous athletes on cards in boxes: 20% of the boxes contain a picture of Hope Solo, 30% of Danica Patrick, and the rest of Blake Griffin. You do not care about completing the collection — you just have to have the Hope Solo card. How many boxes should you expect to open before you find her?

8.1 Bernoulli trials

Assume the pictures are distributed at random and that the manufacturer's claim is true, so the probability of finding Hope in any box is 0.20. Call opening each box a trial, and notice three things.

THE THREE CONDITIONS

1. There are only **two possible outcomes** on each trial, called *success* and *failure*.
2. The **probability of success**, p, is the same on every trial.
3. The **trials are independent**.

Situations like this are called **Bernoulli trials**. We write p for the probability of success and $q = 1 - p$ for the probability of failure; from now on those two letters are reserved.

Coin tosses, inspecting products on an assembly line, and free throws in a basketball game are all Bernoulli trials. The third condition is the one that usually fails in practice, because real populations are finite: sampling without replacement changes the probability for the next trial.

THE 10% CONDITION

Bernoulli trials must be independent. If that assumption is violated, it is still all right to proceed as long as the sample is **smaller than** 10% **of the population**.

8.2 The Geometric model

We want to model how long it takes to achieve the *first* success. The model that does it is the **Geometric probability model**, written Geom(p), and it is completely specified by the single parameter p.

Achieving the first success on trial x requires first experiencing $x - 1$ failures, which gives the formula directly.

GEOMETRIC MODEL FOR BERNOULLI TRIALS: Geom(p)

$X =$ number of trials until the first success occurs.

$$P(X = x) = q^{x-1}p, \qquad \mu = E(X) = \frac{1}{p}, \qquad \sigma = \sqrt{\frac{q}{p^2}}.$$

For the cereal boxes, $P(X = 1) = 0.2$; $P(X = 2) = (0.8)(0.2) = 0.16$; and $P(X = 5) = (0.8)^4(0.2) = 0.08192$. Since Hope is in 1 box in 5, we expect to find her on average in the fifth box, $E(X) = 1/0.2 = 5$ — which is correct, and not easy to prove.

Exercise 1 (Spam)

The security company *Postini* reports that 91% of e-mails are spam. Assume your e-mail is typical, that you use no filter, and that messages are independent. About how many spam messages should you expect to wade through before you find a real one? And what is the probability that the fourth message in your inbox is the first that is not spam?

Solution. **Identify success.** We are waiting for a *real* message, so success is "not spam" and

$$p = 1 - 0.91 = 0.09, \qquad q = 0.91.$$

Let X be the number of messages checked until a real one appears; the model is Geom(0.09).

Expected wait.

$$E(X) = \frac{1}{p} = \frac{1}{0.09} = 11.1 \text{ messages.}$$

The specific probability. For the fourth to be the first success, the first three must fail:

$$P(X = 4) = q^3 p = (0.91)^3(0.09) = 0.0678.$$

On average you check just over 11 messages before finding a real one, and there is slightly less than a 7% chance that the first real message is the fourth one.

$E(X) = 11.1, \qquad P(X = 4) = 0.068$

CAREFUL

Decide which outcome you are calling "success" *before* you write anything down, and make p the probability of that one. Here the interesting event (a real message) is the rare one, $p = 0.09$, not 0.91. Getting this backwards is the most common error with the geometric model.

Exercise 2 (A blood drive)

People with O-negative blood are universal donors; only about 6% of people have it. If donors line up at random,

(a) how many do you expect to examine before finding one who is O-negative?

(b) what is the probability that the first O-negative donor found is among the first four

people in line?

Solution. Check the conditions. Two outcomes: O-negative (success) or not. $p = 0.06$ for each person, because they lined up at random. The trials are not strictly independent — the population is finite — but the donors lined up are fewer than 10% of all possible donors, so the 10% Condition lets us proceed. Let X be the number of donors until one is O-negative; the model is Geom(0.06).

(a)

$$E(X) = \frac{1}{0.06} \approx 16.7 \text{ people.}$$

(b) "Among the first four" means $X = 1, 2, 3$ *or* 4, and these are disjoint, so we add:

$$P(X \le 4) = 0.06 + (0.94)(0.06) + (0.94)^2(0.06) + (0.94)^3(0.06) \approx 0.2193.$$

Blood drives such as this expect to examine an average of 16.7 people to find a universal donor, and about 22% of the time there will be one within the first four in line.

(a) 16.7 (b) 0.219

A SHORTCUT FOR (B)

The complement is quicker and gives the same answer: the first four all fail with probability $(0.94)^4 = 0.7807$, so

$$P(X \le 4) = 1 - (0.94)^4 = 1 - 0.7807 = 0.2193.$$

One line instead of four terms — and the longer the list, the bigger the saving.

8.3 The Binomial model

Now a different question about the same trials. Suppose you buy 5 boxes of cereal: what is the probability of getting *exactly* 2 Hope Solo pictures? Before, we asked how long until the first success; now we want the *number of successes* in a fixed number of trials.

This needs two parameters: the number of trials n and the probability of success p. The model is written Binom(n, p).

The probability of any *particular* arrangement of 2 successes and 3 failures is $(0.2)^2(0.8)^3$. What remains is to count how many arrangements there are. Each different order is called a *combination*, and the number of them is

$${}_nC_k = \binom{n}{k} = \frac{n!}{k!\,(n-k)!}, \qquad \text{where } n! = n \times (n-1) \times \cdots \times 1.$$

For 2 successes in 5 trials,

$${}_5C_2 = \frac{5!}{2!\,3!} = \frac{5 \times 4}{2 \times 1} = 10,$$

so

$$P(\#\text{successes} = 2) = 10\,(0.2)^2(0.8)^3 = 0.2048.$$

BINOMIAL MODEL FOR BERNOULLI TRIALS: $\text{Binom}(n, p)$

$X =$ number of successes in n trials.

$$P(X = x) = {}_nC_x\, p^x q^{n-x}, \qquad \mu = np, \qquad \sigma = \sqrt{npq}.$$

Exercise 3 (Spam in an inbox of 25)

With 91% of e-mails spam, suppose your inbox contains 25 messages. What are the mean and standard deviation of the number of real messages? What is the probability of finding only 1 or 2 real messages?

Solution. Success is again a real message, $p = 0.09$, and now $n = 25$, so the model is $\text{Binom}(25, 0.09)$.

Centre and spread.

$$E(X) = np = 25(0.09) = 2.25, \qquad SD(X) = \sqrt{npq} = \sqrt{25(0.09)(0.91)} = 1.43.$$

The probability. "1 or 2" are disjoint, so add:

$$P(X = 1) = {}_{25}C_1(0.09)^1(0.91)^{24} = 0.2340,$$

$$P(X = 2) = {}_{25}C_2(0.09)^2(0.91)^{23} = 0.2777,$$

$$P(X = 1 \text{ or } 2) = 0.2340 + 0.2777 = 0.5117.$$

Among 25 messages you expect an average of 2.25 real ones, with a standard deviation of 1.43, and there is just over a 50% chance that exactly 1 or 2 of them are real.

$\mu = 2.25,\ \sigma = 1.43,\ P = 0.512$

Exercise 4 (Twenty donors)

Suppose 20 donors come to a blood drive, and 6% of people are universal donors.

(a) What are the mean and standard deviation of the number of universal donors among them?

(b) What is the probability that there are 2 or 3 universal donors?

Solution. The conditions are as before, and the 10% Condition again lets us treat the trials as independent. Let X be the number of O-negative donors among $n = 20$; the model is $\text{Binom}(20, 0.06)$.

(a)

$$E(X) = np = 20(0.06) = 1.2, \qquad SD(X) = \sqrt{20(0.06)(0.94)} \approx 1.06.$$

(b)

$$P(X = 2) = {}_{20}C_2(0.06)^2(0.94)^{18} \approx 0.2246,$$

$$P(X = 3) = {}_{20}C_3(0.06)^3(0.94)^{17} \approx 0.0860,$$

$$P(X = 2 \text{ or } 3) \approx 0.2246 + 0.0860 = 0.3106.$$

In groups of 20 randomly selected donors, expect an average of 1.2 universal donors with a standard deviation of 1.06; about 31% of the time there will be 2 or 3 of them.

$\mu = 1.2,\ \sigma = 1.06,\ P \approx 0.311$

CAREFUL

Geometric or binomial? Ask what the random variable *counts*.
If it counts **trials until the first success**, with no fixed n — how many boxes, how many people in line — it is **geometric**.
If it counts **successes in a fixed number of trials** — how many in 5 boxes, in 25 messages, among 20 donors — it is **binomial**.

8.4 Approximating the Binomial with a Normal model

Suppose the Tennessee Red Cross anticipates needing at least 1850 units of O-negative blood this year and expects to collect from 32,000 donors. How great is the risk of falling short?

In principle we know how: Binom$(32000, 0.06)$. But the probability of getting exactly 1850 units involves ${}_{32000}C_{1850}$, a number with more than 100,000 digits, and the question asks for *at least* 1850, so we would have to do it again for 1851, for 1852, and on to 32,000. No thanks.

Here an old friend comes to the rescue. The binomial model has

$$\mu = np = 32000(0.06) = 1920, \qquad \sigma = \sqrt{npq} = \sqrt{32000(0.06)(0.94)} \approx 42.48,$$

and a Normal model with that same mean and standard deviation turns out to be a very good approximation.

Exercise 5 (Will the Red Cross run short?)

Estimate $P(X < 1850)$.

Solution. Standardize and use the Normal model:

$$P(X < 1850) = P\left(z < \frac{1850 - 1920}{42.48}\right) = P(z < -1.65) \approx 0.05.$$

There is about a 5% chance that this Red Cross chapter will run short of O-negative blood.

≈ 0.05

A Normal model is a close enough approximation only for a large enough number of trials, and what counts as "large enough" depends on p: a very small or very large p needs a bigger sample.

THE SUCCESS/FAILURE CONDITION

A Binomial model is approximately Normal if we expect at least 10 successes and 10

failures:

$$np \geq 10 \quad \text{and} \quad nq \geq 10.$$

Exercise 6 (Is the spam filter too aggressive?)

With 91% of e-mail being spam, you install a filter. Over the past week it okayed only 151 of the 1422 e-mails you received, junking the rest. What is the probability that no more than 151 of 1422 messages are real? Should you worry that the filter is too aggressive?

Solution. **Check the conditions.** The messages are less than 10% of all e-mail traffic. And

$$np = 1422(0.09) = 127.98 \text{ real}, \qquad nq = 1422(0.91) = 1294.02 \text{ spam},$$

both far greater than 10, so the Normal approximation is legitimate.

Compute.

$$\mu = 127.98, \qquad \sigma = \sqrt{1422(0.09)(0.91)} \approx 10.79,$$

$$P(X \leq 151) = P\left(z \leq \frac{151 - 127.98}{10.79}\right) = P(z \leq 2.13) = 0.9834.$$

Interpret. There is over a 98% chance that no more than 151 of the 1422 were real messages — so getting 151 through is entirely unsurprising, and the filter may well be working properly. Nothing here suggests it is too aggressive.

0.983: no cause for concern

Exercise 7 (Dead pixels)

A manufacturer rejects an LCD panel if it has too many dead pixels. When the production line is working correctly, the probability of rejecting a panel is 0.07.

(a) How many rejects are expected in a day's run of 500 screens, and with what standard deviation?

(b) If they reject 40 screens today, is that large enough to be concerned?

(c) Over a week of 5 days they have rejected 200 screens — an average of 40 a day. Should that raise concerns?

Solution. **(a)** With $n = 500$ and $p = 0.07$,

$$\mu = 500(0.07) = 35, \qquad \sigma = \sqrt{500(0.07)(0.93)} = 5.7.$$

(b) Here $np = 35$ and $nq = 465$, both above 10, so we may use the Normal model:

$$P(X \geq 40) = P\left(z \geq \frac{40 - 35}{5.7}\right) = P(z \geq 0.877) \approx 0.19.$$

A day like this happens about one time in five when nothing is wrong. Not an extraordinary number of rejects; no cause for concern.

(c) Now the run is the whole week, $n = 2500$:

$$\mu = 2500(0.07) = 175, \qquad \sigma = \sqrt{2500(0.07)(0.93)} = 12.757,$$

$$P(X \geq 200) = P\left(z \geq \frac{200 - 175}{12.757}\right) = P(z \geq 1.96) \approx 0.025.$$

This would occur by chance only about 2.5% of the time if nothing were wrong. **Yes, that should raise concerns.**

(a) 35 ± 5.7 (b) 0.19, fine (c) 0.025, worrying

WHY ONE DAY IS INNOCENT AND FIVE DAYS ARE NOT

The daily average is identical in (b) and (c) — 40 rejects a day. What changes is n. The mean grows like n while the standard deviation grows only like $\sqrt{n}$, so the same *rate* sits 0.88 standard deviations out over one day and 1.96 over five.
This is the $\sqrt{n}$ of the previous chapter doing its work, and it is the whole logic of statistical inference in miniature: a result that is unremarkable in a small sample becomes convincing evidence in a large one.

8.5 The Poisson model

Siméon Denis Poisson was a French mathematician interested in events with very small probability. He derived his model to approximate the Binomial when the probability of success p is very small and the number of trials n is very large — when, in short, the binomial coefficients become unusable.

The Poisson's parameter is λ, the mean of the distribution. To use it as an approximation to the Binomial we make the means match, setting $\lambda = np$.

POISSON MODEL FOR SUCCESSES: Poisson(λ)

λ = mean number of successes, X = number of successes.

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad E(X) = \lambda, \qquad SD(X) = \sqrt{\lambda}.$$

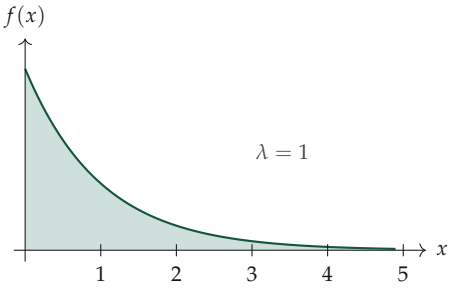
It is a reasonably good approximation to the Binomial when $n \geq 20$ with $p \leq 0.05$, or $n \geq 100$ with $p \leq 0.10$.

Notice what this buys us: no factorials of enormous numbers, and only *one* parameter instead of two. For rare events counted over many opportunities — typing errors per page, accidents per month, website hits per minute — the Poisson is usually the natural model in its own right, not merely an approximation.

8.6 The Exponential model

The Poisson models the *occurrence* of events. The **Exponential probability model**, with the same parameter λ, models the *time between* those events. Because time is continuous, this is a continuous model:

$$f(x) = \lambda e^{-\lambda x}, \qquad x \geq 0,\ \lambda > 0.$$



The shared λ is not a coincidence. If a discrete random variable follows a Poisson model with rate λ, the times between events follow an Exponential model with the same λ. The mean of the Poisson variable is λ; the mean of the Exponential is $1/\lambda$, and so is its standard deviation. That inverse relationship makes sense: if λ increases, we expect *more* hits per minute, so the expected time *between* hits must go down.

Probabilities are areas under the curve, and they take a convenient form:

$$P(s \le X \le t) = e^{-\lambda s} - e^{-\lambda t},$$

and in particular, setting $s = 0$,

$$P(X \le t) = 1 - e^{-\lambda t}.$$

Exercise 8 (Waiting for the next hit)

Hits to a website arrive according to a Poisson model with $\lambda = 4$ per minute. What is the probability of waiting less than 20 seconds for the next hit?

Solution. Work in minutes, since that is the unit of λ: 20 seconds is $t = 1/3$ of a minute. Then

$$P(X \le \tfrac{1}{3}) = 1 - e^{-\lambda t} = 1 - e^{-4/3} = 1 - 0.264 = 0.736.$$

A sanity check: at four hits a minute the arrivals come about every 15 seconds on average, so it is no surprise that nearly 75% of the time we wait less than 20 seconds.

0.736

WHAT CAN GO WRONG

Be sure you really have Bernoulli trials. Two outcomes, constant p, independence — and the 10% Condition when sampling without replacement.
Do not confuse the Geometric and the Binomial. Trials until the first success is geometric; successes in a fixed number of trials is binomial.
Do not use the Normal approximation with small n. It needs at least 10 expected successes *and* 10 expected failures.
Do not assume everything is Normal. Knowing a mean and a standard deviation does not make a model Normal. Using a Normal model where it does not apply leads to wrong answers and misleading conclusions.